

1 Objective-aligned evaluation inflates the distributional gain 2 in single-cell drug retrieval

3 **Running title:** Objective–utility mismatch in single-cell drug retrieval

4 **Authors**

5 **[[Author One]]**¹**[[ORCID]]**, **[[Author Two]]**²**[[ORCID]]**, **[[additional authors]]**, **[[Senior Author]]**^{1,1}

6 **Affiliations**

7 ¹**[[Department, Institution, City, Country]]**

8 ²**[[Department, Institution, City, Country]]**

9 **Corresponding author:** **[[Name]]**, **[[email]]**

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Abstract

Single-cell drug retrieval is moving from mean signatures to distributional scores over treated populations, and routinely reports large gains under a metric of its own distributional kind. We introduce DART, a family of population-to-population scores, and show that mean-signature retrieval is its zero-variance limit, not a competitor. Holding rankings, scorer and queries fixed and changing only the judge, distributional retrieval is decisively ahead under objective-aligned metrics, reverses under mechanism recovery, and beats its incumbent against an external functional oracle, where a scalar comparing no distributions reproduces nearly all of it. Even blind external oracles reverse the winner by statistical form. We formalise when cell-level structure can pay off as two gates, differential response and recoverability; in patient tumours nobody constructed both open, yet no advantage appears: the mean already ranks most of what a subpopulation does. Distributional information here is real, and much smaller than objective-aligned evaluation makes it appear.

Introduction

Single-cell technologies have changed what it means to describe a drug’s effect: it is not a single number but a distribution. Within a treated population, distinct subpopulations can respond in opposite directions, and the state that governs therapeutic outcome is often a minority one, a persister cell or a resistant clone. This premise drives the current wave of investment in virtual-cell models¹, single-cell foundation models^{2–4}, and perturbation-response predictors^{5–7}: if responses can be characterized at single-cell resolution, one should be able to rank candidate drugs better than methods that average the population into a mean signature, as the Connectivity Map does^{8,9}.

That premise is rapidly becoming method. Recent work has brought population-to-population distributional distances, optimal transport, maximum-mean-discrepancy, energy distance and the Wasserstein metric, into single-cell perturbation modeling, and these methods commonly report substantial gains over mean-based approaches. Yet they are almost always scored by distributional metrics of their own kind. When a distributional model is graded by a distributional yardstick, how much of the reported gain is a genuine improvement in decisions, and how much is the method and the yardstick sharing a mathematical target, has been asked too rarely^{10–12}. We name this an objective–utility mismatch, and are deliberate about how strong a word that licenses. Training on an objective and testing with a metric of the same family is not, by itself, a defect: a classifier fit by cross-entropy and scored by held-out log loss is honestly evaluated, because held-out data still tests generalization and the model can still lose. Retrieval is different, and sharper. There is no fitting step and no held-out set; the candidate is chosen by *maximizing a score*, and the evaluation then measures *that same score’s* ground truth on *those same candidates*, so the ranking function and the evaluation function are one function evaluated once, with no estimation error to expose and no way to lose. We reserve the word circular for exactly this condition, and state it formally in Results.

Deciding whether “distributional methods are better” survives past the field’s own metrics requires an oracle that is both independent of the score and *semantically matched to the retrieval task*; getting the second requirement wrong is its own trap, since a similarity retriever graded on absolute potency is being asked a question it was never posed. The distinction we hold throughout is between objective

67 fidelity and independent utility (Fig. 1). Objective-aligned metrics (Class A) legitimately test whether a
68 method computes its own objective; task-proximal metrics such as mechanism-of-action recovery (Class
69 B) are more independent but remain derived from the same upstream data; oracle-independent utility
60 metrics (Class C), such as dose-response viability or longitudinal survival, are what a therapeutic claim
61 ultimately requires, and are largely absent from the benchmarks these methods are evaluated on. An
62 audit across six published method families, spanning signature retrieval^{8,9}, expression prediction^{5,6,13}
63 and graph-based inverse design^{7,14,15}, finds objective-aligned evaluation to be the default in each. The
64 central question is therefore not whether distribution-aware retrieval beats mean retrieval, but where
65 objective fidelity ends and independent utility begins.

66 To make that boundary measurable we introduce DART (Distributional Auditing of Retrieval
67 Transfer), a family of population-to-population retrieval scores. Mean-signature retrieval, including
68 Connectivity-Map cosine, is not a competitor but the zero-variance collapsed member of this same
69 family: under the implemented kernel the two are numerically identical, with Hit@1 agreeing to three
70 decimals. That unification lets us use DART as a controlled probe, changing only the judge while the
71 rankings, the scorer and the queries stay fixed, and watching where the conclusion moves. Under
72 objective-aligned metrics distributional retrieval looks decisively ahead; re-scored for mechanism
73 recovery on the identical rankings the lead does not merely shrink but reverses; against an external
74 functional oracle it does beat its incumbent, yet a scalar that compares no distributions reproduces
75 almost the whole margin. The mismatch even reaches past metrics a method can see: two oracles built
76 from the same proteins in the same cells, differing only in whether each is computed as a mean or as a
77 distribution, hand victory to opposite methods.

78 What bounds the distributional gain is mechanistic, and it is the affirmative payload of this work.
79 A predictor that adds a *single* delta vector to every cell has, by algebra rather than by experiment, two
80 subpopulations with identical responses, so a distributional score has nothing to exploit; usefulness
81 therefore requires that subpopulations both respond *differently* and can be *told apart* at all. We formal-
82 ize this as a two-gate criterion, differential response and recoverability, and test it where it matters: in
83 patient-derived glioblastoma tumours that nobody constructed, rather than in the cell-line mixtures we
84 had assembled, which turned out to distort the biology in both directions at once and to flatter the
85 premise both times. In real tissue both gates open, and the advantage still fails to appear, because the
86 mean already ranks most of what the compartments do. The two gates are necessary, *not* sufficient.

87 Our angle differs from two adjacent lines of work. One provides distributional *prediction* methods,
88 using optimal-transport or Schrödinger-bridge formulations to model perturbation responses^{16,17}; these
89 ask how to predict outcomes better, not how to rank candidate drugs, and do not test whether the evalu-
90 ation is circular. A second compares distributional evaluation metrics for generative models^{18,19}, asking
91 which metric better measures whether a generated distribution matches a real one; we ask the orthogo-
92 nal question of how much of a reported gain is self-fulfilling when the retrieval score and the evaluation
93 metric are the same object. To our knowledge neither this circularity nor the zero-variance identity has
94 been stated before. The work therefore offers this field a test for deciding whether a distributional gain
95 is real or circular, a diagnosis of a failure mode that likely inflates results well beyond drug retrieval^{20,21},
96 and a roadmap of two measurable gates. The roadmap points somewhere specific, and it is the one en-
97 couraging thing we found: in a real tumour the subpopulation information a distributional score would

need is *present*, and it is unsupervised clustering, not the data, that fails to recover it. That is an engineering problem, and engineering problems get solved. As single-cell foundation models proliferate, the question will only become more pressing.

Results

Separating objective fidelity from independent utility requires an apparatus that varies the evaluator while holding everything else fixed. DART provides it: a single family of population-to-population retrieval scores whose candidate rankings can be re-graded by criteria of increasing independence without recomputing the rankings themselves (Fig. 1). The same candidates, ranked once, are scored by DART's own distributional objective (Class A), by a task-proximal biological criterion such as mechanism recovery (Class B), and by an oracle-independent functional readout (Class C), so that from one analysis to the next the rankings, the scorer and the queries are held fixed and only the evaluator changes. Each apparent advantage thereby becomes a testable claim: a distributional gain that reflects better decisions should survive a more independent judge, whereas one that reflects only the scoring of a method with a metric of its own kind should not.

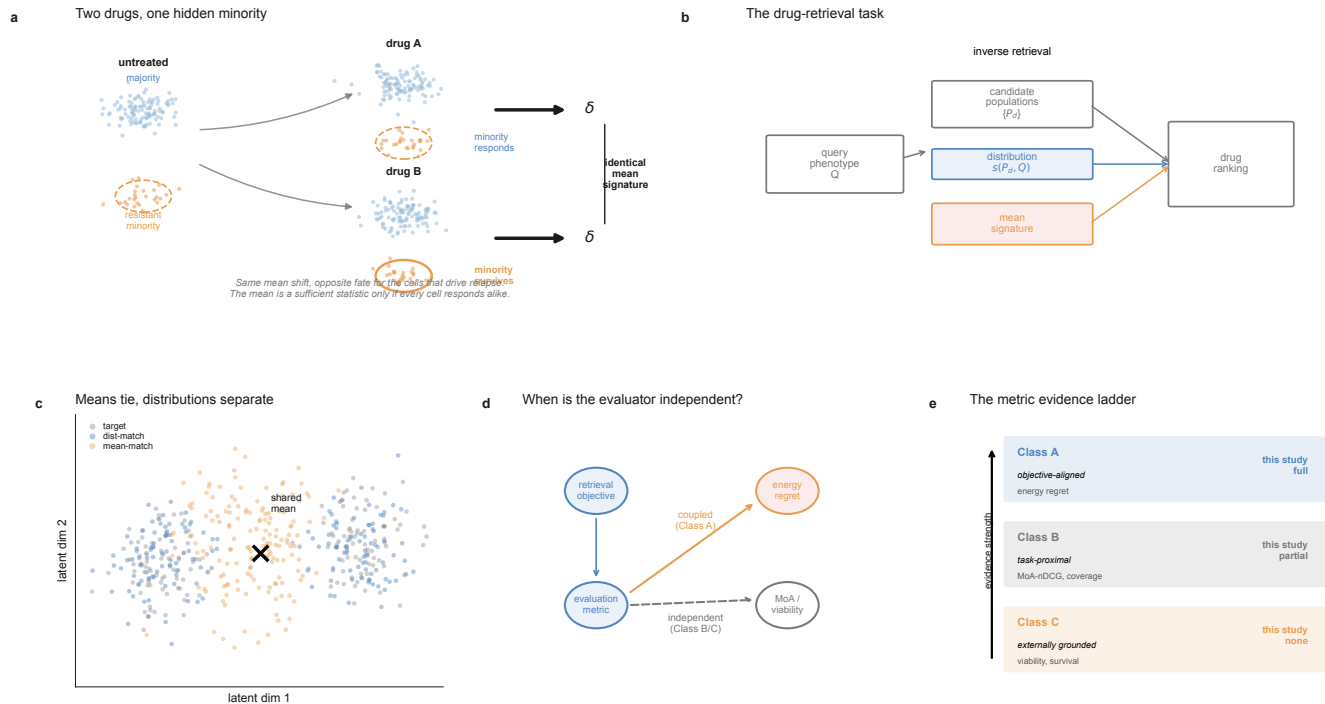


Figure 1: Distributional differences are visible, but their value depends on the evaluator. **a**, Concept: within a treated population, subpopulations respond differently and the outcome-governing state is often a minority one. **b**, Mean-signature retrieval collapses a population to one vector; distributional retrieval keeps the full population. **c**, A target population and candidate library define the inverse retrieval task. **d**, Two candidates can share a mean yet differ in distribution. **e**, Evidence tiers: objective-aligned (Class A), task-proximal (Class B), oracle-independent (Class C). This study reports all three. Its Class-C readout is an imported, semi-real functional oracle built from dose-response profiles in cell lines the transcriptional data never saw; a Class-C claim measured on the same cells would require wet-lab validation.

Mean retrieval is the zero-variance limit of DART

We define the inverse retrieval task at the level of populations. For a target population Q and a library of candidate perturbation populations $\{P_d\}$, a retrieval score $s(P_d, Q)$ ranks candidates by simi-

115 larity of effect. Mean-signature retrieval replaces each population with its mean differential-expression
 116 vector and scores the cosine (or correlation) between mean signatures, the CMap operation. DART in-
 117 stead scores the full populations with a distributional distance (energy, MMD, sliced-Wasserstein, or a
 118 subpopulation-coverage aggregate) and ranks by the negative distance.

119 These are not two unrelated method classes but two ends of one spectrum. Writing a population as
 120 its mean plus a zero-mean residual, the distributional distance reduces exactly to the mean-to-mean dis-
 121 tance when the residual variance is taken to zero. Mean-signature retrieval is therefore the zero-variance
 122 collapsed member of the distributional family. Under the implemented cosine kernel this is not an ap-
 123 proximation but an identity: mean-cosine retrieval and CMap cosine retrieval achieve identical Hit@1
 124 (0.3885, to six figures 0.388492), because they compute the same operation, the cosine of mean-delta sig-
 125 natures. Distributional energy retrieval reaches 0.837 on the same queries. Both figures are unweighted
 126 macro-means over the seven (task $\times$ setting) cells of three retrieval tasks: controlled SciPlex3, cross-line
 127 SciPlex3 and Frangieh.

128 The collapse is one end of a continuous spectrum. The subpopulation-coverage aggregate carries
 129 a temperature β that interpolates between aggregation regimes, and on the controlled benchmark the
 130 interpolated distance rises monotonically from 0.7125 at $\beta \rightarrow 0$, exactly the mean-aggregated value,
 131 to 1.300 at $\beta \rightarrow \infty$, exactly the worst-case value (Fig. 2e). The global energy distance sits inside the
 132 same family, equalling the single-partition ($K = 1$) coverage aggregate exactly (Panobinostat, energy =
 133 coverage $_{K=1} = 0.1099$, difference 0). Energy, coverage and mean retrieval are therefore one axis of vari-
 134 ance sensitivity rather than competing method classes, and mean-signature retrieval is its zero-variance
 135 endpoint (Fig. 2c,f).

136 This unification holds independently of any downstream utility metric: it is an algebraic fact about
 137 the scoring rules, not a claim about which method is therapeutically better. Two scope limits are explicit.
 138 First, the exact identity is specific to the cosine connectivity kernel; the canonical L1000 WTCS score is a
 139 rank-based (GSEA-style) enrichment of the same mean-collapsed signature, monotone in the same quan-
 140 tity but not algebraically identical to DART’s cosine (cmap_wtcs Hit@1 0.4635, which differs). Second,
 141 the identity concerns the scoring operation, not the choice of input population; the value of retaining the
 142 full distribution is exactly what the remaining results interrogate.

143 **Objective-aligned evaluation inflates the gain**

144 The collapse identity above has an empirical face. Across all retrieval metrics, mean cosine and
 145 Connectivity-Map cosine are perfectly correlated ($\rho = 1.00$), while the energy and coverage scores
 146 form a separate, near-orthogonal block (Extended Data Fig. 1, which also gives dataset scale and the
 147 DART metric definitions). The correlation is computed over the 54,180 query–candidate scores that the
 148 1,260 retrieval queries generate against their candidate libraries, at 43 candidates per query. The unit is
 149 the scored pair, not the query: an earlier draft called these “54,180 real queries”, which overstates the
 150 independent sample size by a factor of 43.

151 When candidate responses are *observed*, so that full candidate populations are available rather than
 152 predicted, distributional retrieval looks decisively stronger than mean retrieval. Energy-distance re-
 153 trieval reaches Hit@1 0.837 against 0.389 for mean-cosine and the numerically identical CMap cosine;
 154 PCA-distance retrieval reaches 0.778 against PCA-mean’s 0.518 (Fig. 3). These are unweighted macro-
 155 means over the seven (task $\times$ setting) cells of the three retrieval tasks. Weighting cells by their query

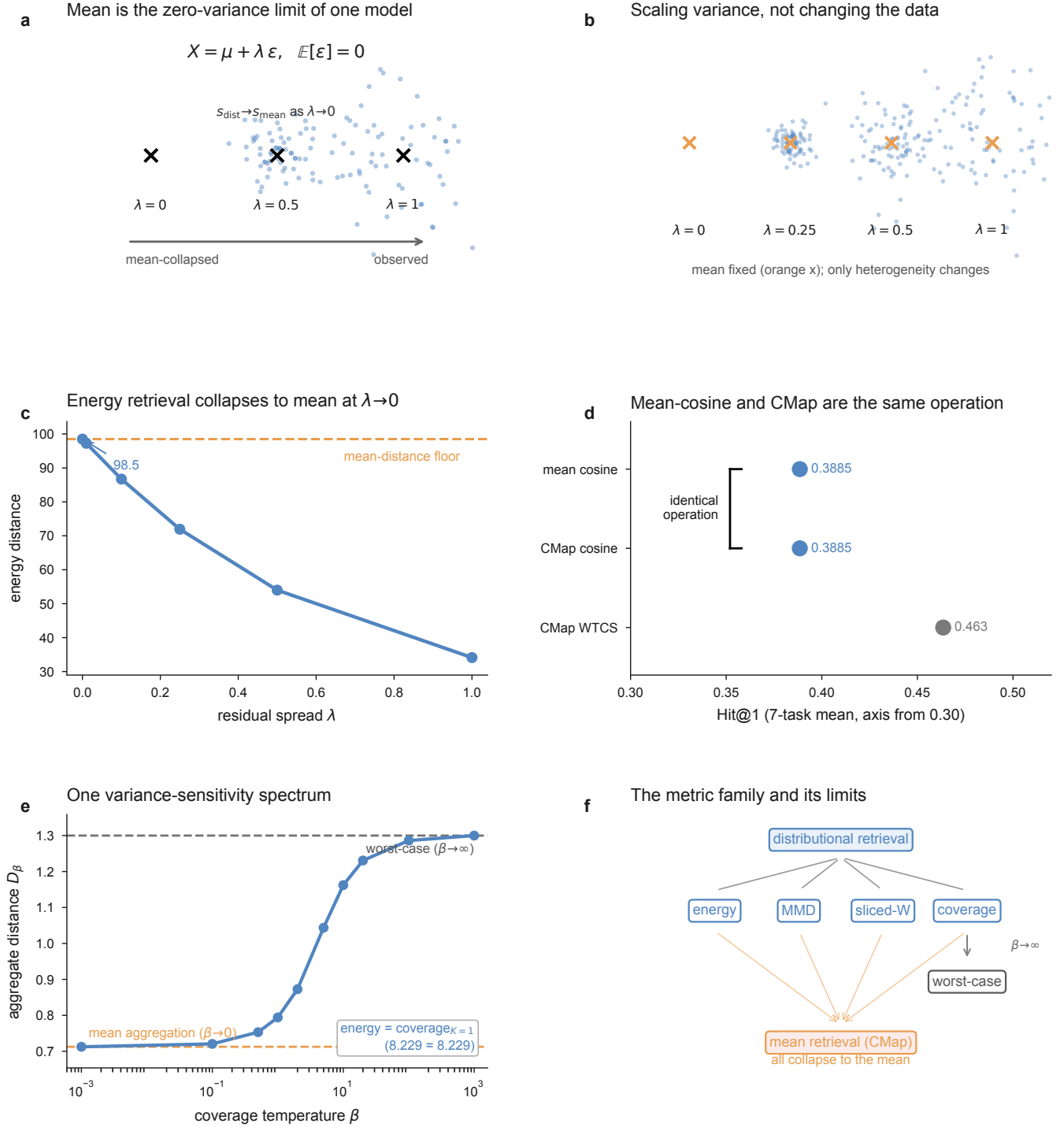


Figure 2: Mean-signature retrieval is the zero-variance limit of distribution-aware retrieval. **a**, Generative model $X = \mu + \lambda \varepsilon$: the mean signature is the $\lambda \rightarrow 0$ limit. **b**, Scaling only residual variance leaves the mean fixed. **c**, Energy distance converges to the mean-distance floor as residual spread vanishes. **d**, Under the implemented cosine kernel, mean cosine and CMap cosine are numerically identical (0.389); WTCS is a rank-based sibling. **e**, The β -interpolation spectrum from mean-aggregation to worst-case, with energy equal to single-cluster coverage. **f**, Metric-family hierarchy: every limit collapses to the mean.

counts gives 0.887 against 0.421; we report the unweighted figure throughout, because it does not let the largest cell dominate.

Extending from observed populations to a partial-observed retrieval setting, we score candidates by an energy-based welfare regret, how much decision regret a ranking incurs against a latent welfare objective. Over all 765 partial-observed queries, the worst-case subpopulation-coverage scorer reduces median regret by +0.118 against mean retrieval (mean +0.258), improving 72% of queries (paired Wilcoxon $p = 1.6 \times 10^{-66}$). We report the full query set here rather than the gate-recommended subset of it: the two are indistinguishable (+0.119 on the 621 recommended queries), and selecting a headline on a gate-selected subset would be exactly the analytic degree of freedom this paper audits.

By the standard of these metrics, distributional retrieval is far ahead. But the metric here is objective-aligned: the energy welfare regret and the DART energy score measure the same distributional object. This result establishes objective fidelity: distributional retrieval computes a discriminative objective, and observed heterogeneous populations contain real distributional signal. It is a *distributional retrieval signal under an objective-aligned proxy*, not evidence of independent therapeutic utility. Whether that signal survives a more independent judge is the question the next sections answer.

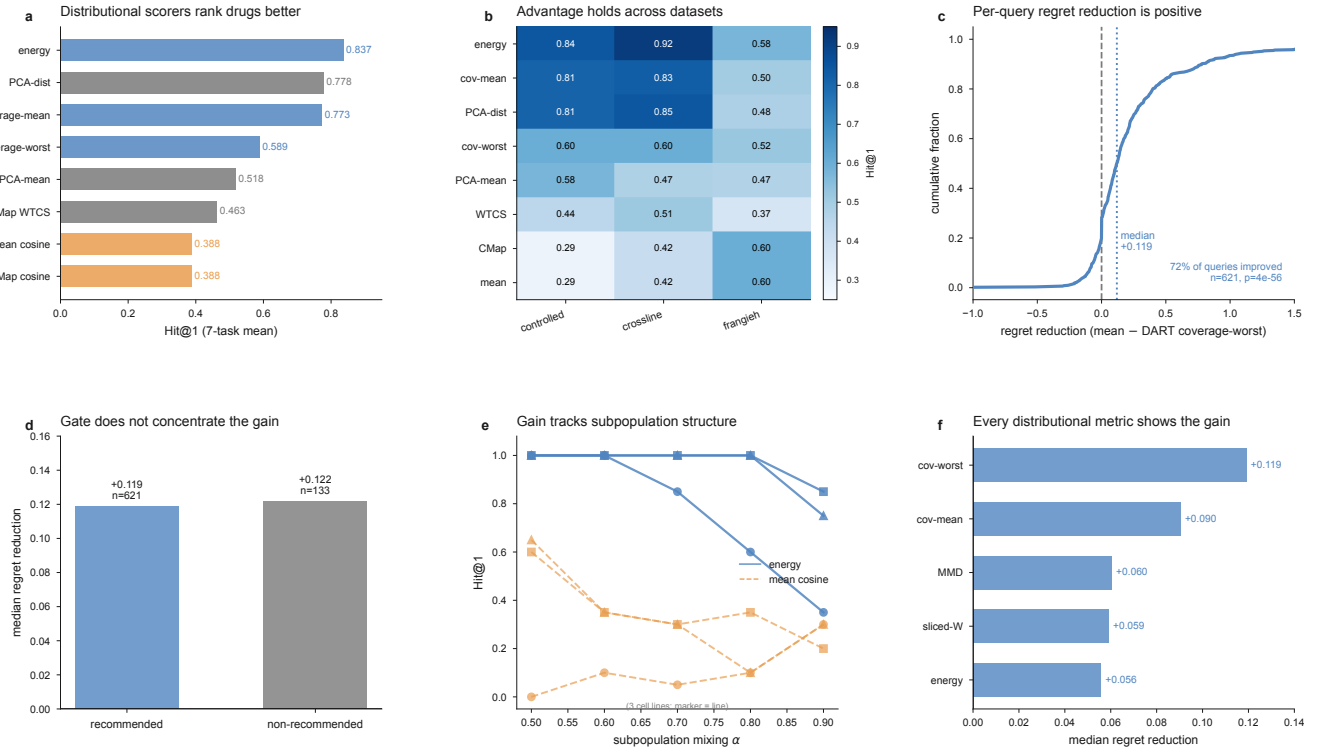


Figure 3: Under objective-aligned metrics, distributional retrieval looks decisively stronger. **a**, Hit@1 ladder: energy 0.837 down to mean/CMap cosine 0.389, unweighted macro-mean over the seven (task $\times$ setting) cells of three retrieval tasks. **b**, Per-task Hit@1. The advantage holds on the controlled and cross-line SciPlex3 tasks but *not* on Frangieh, the only natural real dataset in the macro-mean, where mean/CMap cosine (0.600) beats energy (0.578). The macro-mean hides this counterexample; we state it here and in the text. **c**, Per-query regret reduction over *all* 765 partial-observed queries: median +0.118, 72% improved (the +0.119 quoted elsewhere is the 621-query gate-recommended subset, and the two are indistinguishable, which is panel **d**'s point). **d**, Recommended (+0.119) versus non-recommended (+0.122) subsets: the gate does not concentrate the gain. **e**, Alpha-crossover: the energy advantage narrows as subpopulations merge. **f**, All five DART metrics show positive Class-A regret reduction.

171 Task-proximal metrics collapse the apparent gains

172 This is the central result (Fig. 4). Both sides of the comparison use one scorer on one query set, so the
173 only thing that changes is the class of metric doing the judging. Selecting the best scorer separately for
174 each metric class, on different query subsets, is an analytic degree of freedom this task readily permits,
175 and it would report +0.119 against -0.013: it *understates* the collapse. We fix both and report the larger
176 number.

177 On the 600 partial-observed queries where both metric classes are defined (MoA recovery is unde-
178 fined for the remaining 165; Methods), the worst-case coverage scorer achieves a Class-A regret reduc-
179 tion of +0.129 (median; mean +0.275). Re-scoring the *identical* rankings under mechanism-of-action
180 recovery nDCG, the same two statistics are 0.000 (median) and -0.037 (mean; Fig. 4a). We give both,
181 because they say different things and both are load-bearing: on the median query the advantage does
182 not shrink but vanishes *exactly*, and averaged over queries it turns *negative*. The reversal is not an artefact
183 of a heavy tail: 41.3% of queries *lose* mechanism recovery under the distributional score against 34.2%
184 that gain it (Wilcoxon $p = 2.4 \times 10^{-4}$). DART is not merely no better; it is worse. All five DART variants
185 behave the same way (Class-A gain +0.056 to +0.129, Class-B loss -0.011 to -0.037), so the conclusion
186 does not depend on which variant is chosen.

187 Projecting the real datasets onto the retrieval task, the distributional advantage in minority-state
188 coverage across all 239 real-data tasks is statistically real and practically negligible: mean +0.0018,
189 Wilcoxon $p = 1.3 \times 10^{-7}$ on the 137 tasks with a nonzero difference, on a metric whose observed range
190 is [0.60, 1.00]. In 102 of the 239 tasks the difference is *exactly* zero, because both method families select
191 the same drug. And the effect is confined to the mixtures we constructed: the mean gain is +0.0043 on
192 the 90 constructed cross-line tasks, against +0.0004 within-line ($n = 90$), +0.0004 on Frangieh ($n = 23$),
193 +0.0003 on CD34+ ($n = 12$) and +0.00003 under predicted candidates ($n = 24$) (Fig. 4g).

194 An independent divergence audit reaches the same verdict and numerically the same effect: a cov-
195 erage gain that is real, monotone in divergence, and around +0.002, which is to say too small to act on.
196 Two analyses that share no code path converge on it.

197 **A dominance count on this data counts noise excursions.** The conventional summary would be a
198 count of tasks whose gain clears a threshold, here +0.01. That count is not interpretable (Methods), for
199 two reasons. The threshold sits at 0.77 standard deviations of the nonzero-difference distribution, inside
200 its noise band, and it is one-sided, so 11 tasks clear it for DART while 2 clear it for the mean. And the
201 seed is not a replicate: it re-draws which drugs are held out and re-clusters the minority subpopulation.
202 Holding the drug fixed and resampling the seed ten times, only **2 of the 10** threshold-crossing cross-line
203 tasks remain above it in at least 8 of 10 seeds, while control drugs that clear it at *no* seed in the recorded
204 run clear it in 1 to 4 of 10 resamples. The distribution is what we report.

205 The two survivors are worth naming as a hypothesis rather than a result: A549→MCF7 Abexino-
206 stat (+0.090 ± 0.013, 10 of 10 seeds) and Belinostat (+0.081 ± 0.027, 10 of 10), both HDAC inhibitors,
207 both roughly seven standard deviations above the noise band. With $n = 2$, one cell-line pair and one
208 drug class, this is suggestive and no more, but it is the only place in this study where a distributional
209 advantage on real data survives being checked.

210 Crucially, a diagnostic gate designed to *concentrate* the advantage onto the queries where distribu-
211 tional information should help does not do so. The median energy-proxy regret reduction is +0.119

on the gate-recommended subset ($n = 621$) against +0.122 on the non-recommended subset ($n = 133$; the remaining 11 of the 765 queries are labelled mean-sufficient). The gate does not separate the cases where the distributional score helps from those where it does not, even on the proxy metric it was tuned against, and if anything it selects slightly against them.

The simplest demonstration is at the dataset level. The divergence criterion holds that distributional retrieval can help only when subpopulation responses diverge, operationalized as a response cosine below ≈ 0.9 . Measuring where each dataset actually sits on that axis, cross-line SciPlex3 lies at cosine 0.032, Frangieh at 0.709, and CD34+ at 0.186. All three clear the gate. Yet DART wins only on cross-line: on Frangieh, the only natural real dataset, the mean-signature incumbent wins (Hit@1 0.600 versus 0.578), and CD34+ is our *negative control*, where the mean wins decisively. A criterion that admits the positive anchor and the negative control alike is not a criterion, and the dataset-level divergence rule therefore fails on its own terms.

A direct audit makes the failure concrete (Fig. 4e,f). The gate’s structure-reliability axis is anti-correlated with true response divergence (Spearman $\rho = -0.21$, $p = 3.8 \times 10^{-9}$), moving opposite to the quantity it is meant to track; its preference-conflict axis is only weakly aligned ($\rho = +0.18$); and the recommendation label does not separate high- from low-divergence queries (Mann-Whitney $p = 0.090$). Bypassing the gate and stratifying directly by true divergence, the highest-divergence stratum shows a statistically significant but negligible minority-coverage gain (median gap rising from +0.0009 in the lowest quartile to +0.0018 in the highest, all Benjamini-Hochberg $q \approx 0$, against a metric operating range of $[0.89, 0.99]$: about 0.13 standard deviations). Mechanism recovery is null in the highest stratum ($q = 0.57$, $n = 143$) and *significantly negative* in the lowest (median gap -0.030 , $q = 5.8 \times 10^{-5}$, $n = 166$): where responses are least divergent, DART actively degrades annotation recovery. The deficit is in the data and the task, not only in the gate.

The obvious objection is that a null is merely an underpowered study, and the two metric classes are indeed unequally powered (Extended Data Fig. 2): the coverage metric is fully powered (achieved power 1.00), while MoA recovery is not (0.06; 80% power would need roughly 20,800 queries against the 143 available). We therefore do *not* claim to have proven the absence of a small MoA effect. We claim something weaker and sufficient: where the theory predicts the largest gain there is no detectable gain, and where responses are least divergent the effect that *is* detectable runs against DART.

The conclusion is deliberately strong and deliberately bounded: apparent distributional gains are metric-dependent and cannot be interpreted as therapeutic utility without independent validation. One scorer, one query set and one set of rankings is “far ahead” or “worse than the incumbent” depending only on which metric class judges it. That dependence, not a verdict on any single method, is the finding.

A functional oracle shows the gain is mostly magnitude

Class B is task-proximal, not independent, so the ladder is incomplete without an external functional readout. Choosing that readout is where a Class-C analysis is most easily got wrong, and the wrong choice is the tempting one.

The oracle must match the task. A retrieval score is asked which candidate’s response distribution most resembles the query’s. It is *not* asked which candidate is most cytotoxic. Grading a similarity retriever on absolute potency therefore tests a question it was never posed: handed a weak query, a correctly working retriever *should* return other weak drugs. We report that mismatched comparison

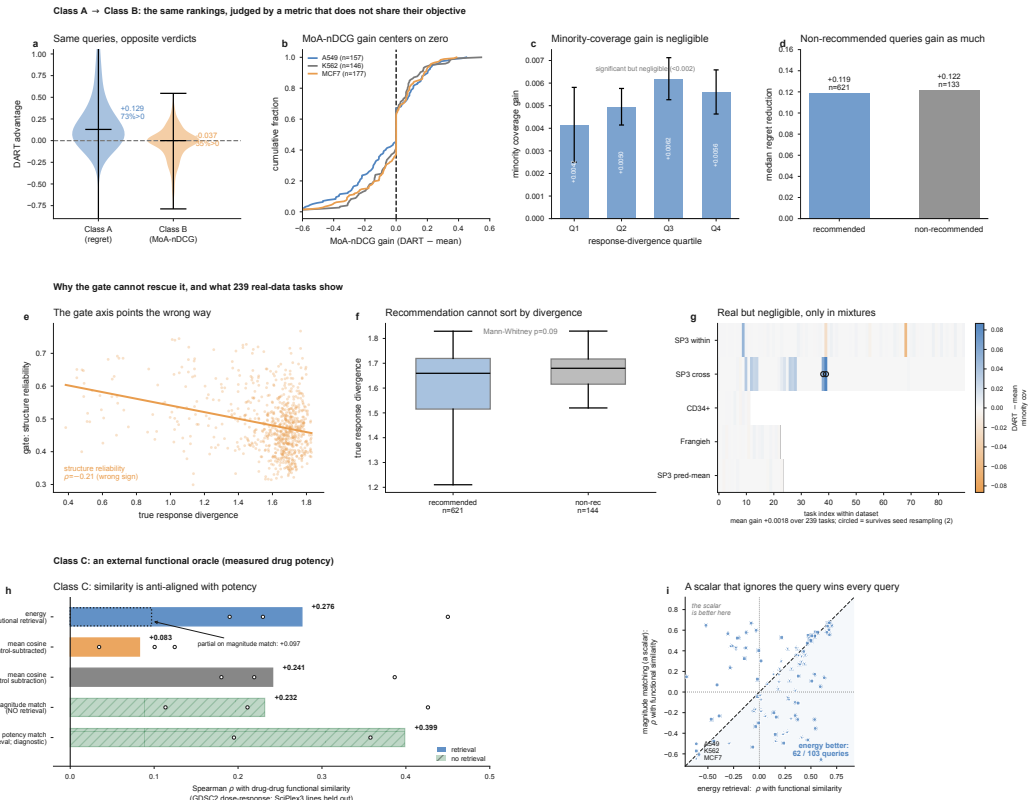


Figure 4: The apparent gains do not survive independent evaluation. The metric-class ladder, read row by row. **Row 1, Class A $\rightarrow$ Class B.** **a**, One scorer, one query set, only the class of metric changes: a Class-A regret reduction of +0.129 becomes a Class-B MoA-nDCG gain of -0.037 . **b**, MoA-nDCG gain by cell line, uniformly at or below zero. **c**, Minority-coverage gain by divergence quartile: significant, and negligible. **d**, The gate does not concentrate the advantage (+0.119 on 621 recommended queries versus +0.122 on 133 non-recommended). **Row 2, why the gate cannot rescue it.** **e**, The gate's structure-reliability axis is *anti*-correlated with true response divergence ($\rho = -0.21$), moving opposite to the quantity it is meant to track. **f**, Recommended and non-recommended queries have indistinguishable divergence (Mann-Whitney $p = 0.09$). **g**, The real-data coverage advantage, task by task, across all 239 tasks: real but negligible (mean +0.0018) and confined to the constructed cell-line mixtures (+0.0043, against +0.0003 in the natural data and +0.00003 under predicted candidates). We plot the difference rather than a count of “dominant” tasks: the conventional 0.01 threshold sits at 0.77 s.d. of the nonzero-difference distribution, inside its noise band, and under seed resampling only 2 of the 10 threshold-crossing cross-line tasks survive (circled: A549 $\rightarrow$ MCF7 Abexinostat and Belinostat, both HDAC inhibitors, above threshold in 10 of 10 seeds). **Row 3, Class C, an independent functional oracle.** 103 leave-one-drug-out queries, SciPlex3 at 10 μ M. The oracle is *drug-drug functional similarity*, not absolute potency: the rank correlation of two compounds' GDSC2 dose-response AUC profiles across the 966 cell lines left once the three SciPlex3 lines are excluded, so it is measured out of context (Methods). **h**, Energy reaches $\rho = +0.276$ against +0.083 for the correctly specified mean incumbent: the first oracle-invisible criterion here on which the distributional score wins. It is not a *distributional* win. A query-dependent scalar matching only the *size* of a candidate's response reaches +0.232 (hatched), and partialling that channel out leaves energy with +0.097 (dotted). Open circles, per-cell-line medians. **i**, Per query, energy against that scalar. The cloud straddles the diagonal: energy wins 62 of 103 queries, and the nominal $p = 0.073$ is anticonservative because queries within a line share a candidate pool. Scored against *absolute* potency instead, every ranking inverts and the magnitude scalar wins outright; that comparison poses a question a similarity retriever was never asked and is reported only as a confounder audit (Extended Data Fig. 5).

only as a confounder audit (below), because the conclusion it invites, that retrieval is anti-aligned with therapeutic value, does not follow from it.

The oracle that does match the task is drug–drug functional similarity: do the query and the candidate *behave* alike in an independent assay? GDSC2 supplies this directly. Each compound has a dose-response AUC profile across roughly a thousand cell lines, and two drugs with similar cross-cell-line sensitivity patterns are functionally similar however potent either happens to be. We define $\text{oracle}(q, c) = \text{Spearman}(\text{AUC}_\bullet(q), \text{AUC}_\bullet(c))$ over the 966 GDSC2 cell lines that remain after *excluding* the three SciPlex3 lines, so the oracle is never evaluated in the context the transcriptional query came from. Because a correlation is centered per drug, a compound’s overall potency level cannot drive it; only its selectivity *pattern* can. Cell lines differ enormously in general drug sensitivity, and left uncorrected that shared main effect makes almost every drug pair look alike (85% of drug pairs positive, median +0.178, versus 52% and +0.015 once corrected); we therefore subtract each cell line’s mean AUC, estimated across all 286 GDSC2 compounds rather than only the matched ones, before correlating.

That correction decides the result, and we report the dependence rather than bury it. Without it, energy falls to +0.097 and drops *below* the mean-signature incumbent’s +0.138, reversing the one affirmative finding of this section; the magnitude scalar (+0.217) beats both. Both oracles are given in full in Supplementary Table 4. The correction is defensible on its face, it was specified on the reasoning above before the retrieval results were inspected, and leaving a shared cell-line main effect in an oracle would be an error of its own. But it is an analytic choice, and a paper arguing that the choice of criterion decides the winner cannot exempt its own criterion from that finding. The honest statement is that the Class-C positive below holds under the oracle we argue is correct and not under the oracle we argue is wrong.

On this oracle, distributional retrieval beats its incumbent (Fig. 4h). Across 103 leave-one-drug-out queries, energy retrieval attains $\rho = +0.276$, against +0.083 for the correctly specified mean-signature incumbent: a threefold margin, in the same direction in all three cell lines. This is the first point in this study at which the distributional score beats its incumbent on a criterion no scorer can see, and we report it as such.

It is nevertheless not a distributional result. Rank the candidates instead by how closely their response *magnitude* matches the query’s, a query-dependent scalar that compares no distributions at all, and it reaches $\rho = +0.232$. Energy’s edge over that scalar is 62 of 103 queries, which even under an anticonservative paired test is not significant ($p = 0.073$). Partial out the magnitude channel and energy retains +0.097 (per line +0.131, +0.053, +0.162: positive in all three, but small).

Even a *query-independent* ordering carries comparable alignment: ranking candidates by their own response magnitude, an ordering identical for every query that performs no retrieval whatsoever, gives $\rho = -0.330$, so the same fixed ordering reversed gives +0.330. We report its magnitude rather than calling it a better recommender, and deliberately so: its *sign* would have to be known in advance, and a paired comparison in fact favours energy on 65 of 103 queries even though its marginal median is the larger of the two. The point is only that a constant ordering is not far behind a distributional score on this oracle. Finally, a single scalar matching the candidate’s potency to the query’s reaches +0.399, above the distributional score, though it is a diagnostic rather than a usable recommender because it requires the query drug’s own measured potency; what it establishes is that the functional oracle retains a substantial

294 potency-level axis, which is the channel the mismatched comparison below confounds entirely.

295 The picture is therefore layered rather than negative. Distributional retrieval carries genuine func-
296 tional signal that the mean signature does not; almost all of that signal is a response-magnitude channel
297 that a scalar reproduces; and the residual attributable to the *distribution* is roughly +0.097, about a third
298 of energy's raw +0.276 and positive but small. We deliberately do not express this as a ratio against
299 the Class-A gain: a Spearman correlation against an external oracle and a Hit@1 or regret improvement
300 against an objective-aligned proxy are different quantities on different scales, and dividing one by the
301 other would manufacture a number rather than measure one. The comparison that is legitimate is within
302 the oracle, and there two thirds of the distributional score's advantage is magnitude. Objective fidelity
303 does not make the distributional gain imaginary. It makes it look far larger than it is.

304 Two operational lessons follow. First, *the magnitude control belongs in every distributional retrieval paper*:
305 rank the candidates by the size of their own response, look at nothing else, and show that you beat it. On
306 the semantically matched oracle we do not beat it significantly, and on the mismatched one we lose to
307 it outright. Second, a mean-signature baseline scored *without* control subtraction reaches +0.241, close
308 to energy and far above the correctly specified incumbent's +0.083, because it measures baseline cell
309 identity rather than drug response. A study using it as its incumbent is comparing against a broken
310 baseline and will overstate its own margin.

311 **Confounder audit: why the mismatched oracle inverts.** Scored against *absolute* potency, energy
312 retrieval is anti-correlated ($\rho = -0.520$, same sign in all three lines) and the query-independent magni-
313 tude scalar reaches +0.692. This is arithmetic, not pathology. Response magnitude predicts potency on
314 its own ($\rho = -0.58$ to -0.71 by line), and the energy *distance* between query and candidate is largely
315 a function of the candidate's own magnitude ($\rho = +0.791$), because big-response populations sit far
316 from everything. Ranking nearest-first therefore returns the smallest-response, weakest candidates. The
317 finding is that the same magnitude channel drives both oracles; it is not evidence that retrieval opposes
318 therapeutic value, and we do not claim that it is.

319 **Scope.** This oracle is semi-real. Viability is measured in a separate bulk assay, joined by compound
320 identity rather than shared cells, under a different dose regime, so it constrains what a transcriptional
321 score recovers about drug function without being a clinical endpoint. The 103 queries within a cell line
322 share a candidate pool, so paired tests across them are anticonservative and are reported as such; the
323 honest unit is the three cell lines, and we give per-line values throughout. The strongest limit on this
324 section is the one stated above: the affirmative result is not robust to the cell-line centering. Under the
325 uncorrected oracle energy reaches only +0.097, below both the mean-signature incumbent (+0.138) and
326 the magnitude scalar (+0.217), so the claim that a distributional score beats its incumbent on an external
327 criterion holds under one construction of that criterion and not the other. What survives both construc-
328 tions is the negative: under neither oracle does the distributional score beat the scalar that compares no
329 distributions.

330 Oracle shape not biology decides which method wins

331 The Class-C result above is not the only external readout available, and the second one disagrees with it.
332 Frangieh Perturb-CITE-seq measures 20 surface proteins in the *same* 218,331 cells whose RNA the scorers
333 read, giving an oracle that is independent of the transcriptome, invisible to every scorer, and free of the
334 cross-assay join the GDSC oracle needs. Scored against it, the *mean* beats energy (+0.242 against +0.146),

335 in all three immune conditions. Taken at face value, one external oracle says distributional retrieval wins
336 and another says it loses.

337 They are not in conflict. Look at what the protein oracle is: cos of the two knockouts' *mean* protein
338 deltas. It collapses the protein readout to a mean, and the mean-cosine RNA scorer computes that same
339 statistic in the other modality. A mean-shaped oracle is structurally matched to a mean-shaped scorer,
340 and energy is being graded on a criterion that discarded the very thing it compares. That is the mirror
341 image of the Class-A alignment this paper is about, arising now between two *external* criteria, and it is
342 testable directly.

343 **We therefore rebuilt the oracle from the same cells and the same proteins, twice, changing only its**
344 **statistical form** (Fig. 5c; Methods). One version is the mean cosine above; the other is the energy distance
345 between the two knockouts' protein-response *cell clouds*, both referred to that condition's control cells so
346 it measures response and not baseline identity. The two RNA rankings under test are byte-identical
347 across the two columns. Nothing about the scorers, the cells, the proteins or the queries changes. Only
348 the shape of the judge changes.

349 **The winner swaps.** Under the mean-shaped oracle the incumbent leads (+0.242 against +0.146); un-
350 der the distribution-shaped oracle the distributional score leads, and by more (+0.529 against +0.334).
351 The reversal holds independently in each of the three immune conditions (Control, IFN γ and Co-culture;
352 $n = 213, 227$ and 219 queries). No scorer can see either oracle, so this is not circularity in the Class-A
353 sense of a method grading its own objective. It is something we did not anticipate and consider the
354 sharpest result in this paper: even when the criterion is external, independent and blind, its own statisti-
355 cal form selects the winner. A field that validates distributional methods against distributional readouts,
356 and mean-based methods against mean readouts, will find each vindicated, and neither finding will be
357 about biology.

358 **The magnitude control survives this, and refines it.** An energy distance tracks a candidate's
359 own response magnitude ($\rho = +0.791$), and both distributional objects here are energy distances, so
360 a magnitude-to-magnitude channel could in principle produce the swap with no distribution ever
361 being compared. It does not, quite. The magnitude scalar does climb steeply when the oracle turns
362 distributional (from +0.125 to +0.352), which is the confound behaving exactly as predicted and is itself
363 worth reporting. But energy still leads that scalar by +0.177 under the distributional oracle, against
364 only +0.022 under the mean-shaped one. So there is a genuine distributional signal in these cells, and
365 *it is visible only to a criterion that is itself distributional*. Whether that makes it a real advantage or an
366 unfalsifiable one is the question this paper leaves open, and we prefer to leave it open than to answer it
367 with a criterion of our own choosing.

368 **The information condition is a two-gate criterion**

369 The preceding results pose a mechanistic question: why is the distributional signal large on observed
370 populations yet absent under independent metrics? The answer is a *candidate-response information con-*
371 *dition* with two distinct gates. A distributional score can only exploit subpopulation structure whose
372 subpopulations (i) actually respond *differently* in the candidate populations supplied to the retriever,
373 and (ii) can be told apart from those populations at all. Both gates are closed *in the constructed cell-line*
374 *data this section uses*, which is why the observed-population result is an upper bound rather than an op-
375 erating point there. That qualifier is load-bearing and we did not originally have it: on natural patient

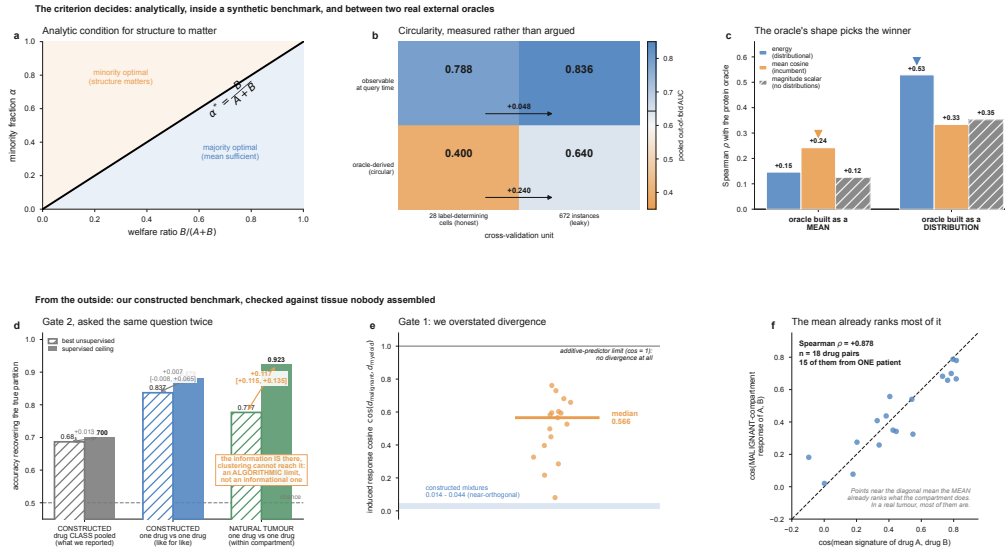


Figure 5: The criterion decides the answer, from the inside and from the outside. Row 1, the criterion decides. a, HIR-Bench's analytic flip boundary $\alpha^* = B / (A + B)$ separates the regime in which the mean is a sufficient statistic from the one in which subpopulation structure can change the decision. **b,** Circularity measured rather than argued, on the 13,440-instance grid: two feature sets $\times$ two cross-validation units. Features a method can *observe* at query time reach AUC 0.788 against a 0.643 majority-class rate (the rule on the colour bar); features derived from the benchmark's own latent utility matrix, of which the label is also a function, are no better than chance (0.400). Arrows give the interaction: pseudo-replication inflates the circular feature set by +0.240 and the honest one by only +0.048. Effective n is 28 parameter cells, not 13,440, and every fold is single-class, so the 0.400-versus-0.5 gap is not itself interpretable. HIR-Bench's phase boundary is designed into its generator and its transfer to real data fails (Supplementary Note 1); its remaining panels are in Extended Data Fig. 6. **c,** The same effect between two *real, external* criteria. An oracle is built twice from the same 20 surface proteins in the same 218,331 cells, changing only its statistical form (mean protein delta cosine, or energy distance between protein-response cell clouds), and the same two RNA rankings are scored against both. **The winner swaps:** the mean incumbent leads under the mean-shaped oracle (+0.242 against +0.146), the distributional score under the distribution-shaped one (+0.529 against +0.334), independently in each of the three immune conditions. No scorer sees protein data, so alignment operates here between external, blind criteria. Hatched, the mandatory control: a query-dependent scalar matching only response magnitude, which compares no distributions. It climbs from +0.125 to +0.352 as the oracle turns distributional, the known magnitude channel ($\rho = +0.791$) behaving as predicted, yet energy still leads it by +0.177 there against +0.022 under the mean-shaped oracle. **Row 2, from the outside.** The two gates on ZhaoSims2021, slice culture and biopsy from ten glioblastoma patients (validated marker assignment; Methods, Supplementary Table 3). **d,** Gate 2, asked the same question in both settings: one drug against another *within one compartment of one patient* (36 splits, 4 patients), against the like-for-like constructed split. Bars are marginal medians; the annotated gap is the median *paired* difference, the correct and smaller statistic. The constructed ceiling of 0.692 was largely an artefact of *pooling* drugs into classes (grey): one drug against one drug, the same cells give 0.879. In a real tumour the information is present (ceiling 0.923) but unsupervised clustering cannot reach it (0.777; gap +0.117, 95% CI [+0.115, +0.135] from a patient-level cluster bootstrap, against +0.007 [-0.008, +0.065] constructed; k -means 0.638, Leiden 0.780). The gap does not depend on the maximisation: with every fixed probe-clusterer pair and no selection anywhere it is +0.122 in the tumour against -0.002 constructed. Dropping the patient that supplies 30 of the 36 splits leaves +0.123 [+0.098, +0.138] on the other three. **Gate 2 in real tissue is an algorithmic bottleneck, not an informational one,** contradicting both our original claim and the cell-type comparison (0.964) with which we first tried to correct it; both are withdrawn. 30 of the 36 splits come from one patient and 39.9% of cells are dropped by the assignment margin, so both numbers are optimistic. **e,** Gate 1: induced response cosine, median 0.566 over 17 patient-drug pairs, against 0.014–0.044 in our constructed mixtures. On a divergence scale ($1 - \cos$: 0.99 against 0.43) the overstatement is roughly twofold; the ratio of the cosines is not a meaningful measure of it. **f,** The premise: over 18 within-patient drug pairs, mean-signature similarity ranks malignant-compartment response similarity at Spearman $\rho = +0.878$. *No p-value is given:* 15 of the 18 pairs come from one patient and reuse the same six drugs, so a test treating them as independent would be anticonservative. Both gates open here and the distributional advantage still does not appear: the two gates are necessary, not sufficient.

376 tumours both gates turn out to be open, and the last Results section is the one that corrected us.

377 **Gate 1, differential response.** We test what generated candidate populations actually contain, using
378 predict-then-rank experiments in which candidate responses are produced by a perturbation predictor
379 rather than observed (Fig. 6). Across three predictor families the distributional advantage over mean
380 retrieval is small and *not consistent in sign*: the nDCG@10 delta is -0.016 (average-effect), -0.071 (nearest-
381 neighbour) and $+0.027$ (CPA-style linear-latent, a surrogate for the latent-arithmetic family; we do not
382 benchmark a trained scGen). DART also wins on Hit@1 for two of the three. We therefore do not claim
383 the advantage is uniformly absent under prediction, only that it is small, unstable, and does not survive
384 the independent metrics of the previous sections.

385 The structural explanation is not the intuitive one. Predicted candidate populations do *not* collapse
386 subpopulation structure. Synthesizing a predicted population by applying the predicted effect to the
387 query context’s real control cells, which is what the latent-arithmetic models in this class actually do,
388 the subpopulation-variance ratio of predicted populations is 0.142 against 0.138 for the real blended
389 candidate population that a scorer ranks; the latent model even sharpens it, to 0.462 (Fig. 6c). Structure
390 survives.

391 What does not survive is *divergence*, though the way we first measured it was doubly wrong and
392 the corrected version says something sharper. Measuring the induced response cosine between the two
393 subpopulations of a candidate population, $\cos(d_{\text{maj}}, d_{\text{min}})$ with each subpopulation referred to its own
394 matched control, real candidate populations respond near-orthogonally (**0.014**) while predicted ones re-
395 spond in largely aligned directions (0.186 average-effect, 0.239 linear-latent, 0.366 nearest-neighbour).
396 Those predicted values do not mean what we took them to mean, for two reasons given below: the
397 populations are cross-context blends, and for a latent model the quantity is contaminated by the genera-
398 tor’s reconstruction error. We retain them because the structure-preservation conclusion above does not
399 depend on them, and we do not build any claim on them.

400 The limit of this behaviour is algebraic, not empirical, and the algebra is what carries the argument.
401 Call a predictor additive if it adds a *single* delta vector to *every* cell of the query context, whether in gene
402 space or through an invertible latent map. Its two subpopulations then have *identical* response deltas,
403 $d_{\text{maj}} = d_{\text{min}} = \delta$, so their induced cosine is exactly 1 and the divergence a distributional score could
404 exploit is exactly zero by construction. No amount of training changes this. Additivity is thus a *sufficient*
405 condition for Gate 1 to be shut, and it is a property of a model class, not a claim about any particular
406 software.

407 Which deployed predictors fall in that class is a separate, empirical question, and the within-context
408 measurement can now answer part of it. Every number below is the model’s *true* induced divergence,
409 taken against its own baseline, and every model is first required to pass a learning check: the cosine
410 between its predicted mean delta and the true one. A model that has not learned the drug effect cannot
411 be said to open or shut any gate, and its divergence is noise structure. We enforce that rather than assume
412 it.

413 Real treated cells, split by the same control-derived cell state, give the target: their two subpopula-
414 tions respond at a cosine of 0.205, so within-context subpopulations in real data do respond very differ-
415 ently. Against that, the average-effect predictor (learning check 0.502) and the linear-latent model (0.343)
416 both return 1.000000 exactly, reproducing none of it, as the algebra requires. The published scGen vari-

417 ational autoencoder (learning check 0.420), which this paper previously declined to say anything about,
418 returns 0.972: its nonlinear decoder buys a small departure from strict additivity, and it reproduces
419 roughly 3% of the divergence real cells show. *A deployed latent-arithmetic model is therefore near-additive in*
420 *practice, not only in principle*, and Gate 1 is essentially shut for it.

421 **A genuinely non-additive predictor does open Gate 1, but only a crack.** A predictor whose dis-
422 placement depends on the cell it is applied to should, if the two-gate account is right, show a divergence
423 below the additive ceiling of 1. We tested an entropic optimal-transport map, the non-parametric core
424 of the neural-OT perturbation models^{16,17}. Run naively in gene space it failed its learning check (cosine
425 0.057 between predicted and true mean delta), a failure of the curse of dimensionality and a cross-context
426 baseline confound rather than of the idea; run in a latent space, as every deployed neural-OT model is,
427 and with each fit context centred on its own control, it passes (0.352, above the 0.30 bar; the configuration
428 was fixed by this learning check, not by the divergence it is used to measure). Its *true* induced divergence
429 is then 0.900 ± 0.101 , significantly below the additive ceiling of 1 (about five standard errors below the
430 mean over $n = 30$), so a non-additive predictor genuinely gives different subpopulations different re-
431 sponses. But real treated cells sit at 0.205, so the map closes only about an eighth of the distance from
432 the additive ceiling to what biology shows, and its learning is itself weak, so part even of that crack is
433 prediction noise.

434 The published CPA model behaves differently and no better: it reconstructs cells acceptably ($R^2 \approx$
435 0.39) but does not learn the drug-specific effect, its predicted mean delta correlating with the true one at
436 only 0.08, so it is below the learning bar and its induced divergence is not reportable. Raising the per-
437 drug training data fivefold, from sixty to three hundred cells, leaves this unchanged (learning 0.08 either
438 way): with 189 drugs the adversarial objective holds the per-drug signal near chance, and more cells
439 do not rescue it. Nor does conditioning its perturbation embedding on the same 2048-bit fingerprints
440 in place of a freely learned vector (learning 0.03); the failure is the adversarial objective, not the drug
441 representation.

442 A modern flow-matching predictor meets the same wall from the opposite side, and shows why.
443 CellFlow²², an optimal-transport flow-matching model conditioned on 2048-bit Morgan (ECFP4) finger-
444 prints of each compound, learns drug responses when trained on all three cell lines (mean-delta cosine
445 0.24 over twenty drugs, reaching 0.81 for the HDAC inhibitor AR-42, though the twenty-drug average
446 sits below our 0.30 bar) and opens Gate 1 (induced divergence 0.85). But the clean test requires pre-
447 diction into a held-out context, and there its learning collapses to 0.03 at the identical training budget
448 and drug set, an $8.4\times$ drop. This holds beyond the mean: re-scored with gene-resolved and distribu-
449 tional metrics, the held-out-context predictions match the true differential-expression direction only at
450 chance (0.52), recover 7% of the true differential genes, and sit on average farther from the true treated
451 population than the unperturbed control does (energy ratio 2.1). Within context the same metrics are
452 informative (0.70 direction, energy ratio 0.98), so the collapse is a real loss of the response, not an artefact
453 of the mean-based check. The collapse is mechanistic: per drug, leave-one-context-out learning tracks
454 that drug’s cross-context response similarity (Pearson $r = 0.85$), so a predictor conditioned on chemistry
455 alone recovers only the context-shared part of a response. On this dataset that shared part is small (me-
456 dian cross-context cosine 0.04) and near zero for exactly the response-divergent drugs the retrieval task
457 selects.

458 None of the three non-additive predictors we could run therefore serves as a clean test: the optimal-
459 transport map and the flow-matching model open Gate 1 but do not carry their learning across contexts,
460 and CPA reconstructs cells but not responses.

461 **We then asked the decisive question, does opening Gate 1 this far buy any retrieval gain, and**
462 **could not answer it, for an instructive reason.** Re-running the cross-line predict-then-rank with the
463 OT map as the candidate-response source, its predictions are too poor to retrieve with at all: Hit@1 is
464 0.05 under *every* rule, mean cosine included, against 0.88 for the additive average-effect predictor. The
465 leave-two-contexts-out transfer this retrieval task demands is harder than the within-context divergence
466 measurement, and the same weak learning that lets the OT map only crack Gate 1 open leaves it near
467 chance as a predictor here. We therefore lack a predictor that both predicts accurately and opens Gate 1,
468 which is exactly what a clean Part-B test needs, so the question stays open. That we could not build one
469 is consistent with the rest of this paper, in which the accurate part of every predictor is its mean and the
470 part that departs from the mean is noise, but it is a limitation rather than a falsification and we present it
471 as such.

472 **Why the measured values are not 1, and what we got wrong about that twice.** An earlier version
473 of this paper attributed the shortfall to the second gate, arguing that a scorer must estimate the partition
474 and does so imperfectly. That is incorrect, and the experiment’s own calibration check exposed it: an
475 additive predictor is required by the algebra above to return exactly 1.000, and it returned 0.267. When a
476 control that is additive *by construction* misses its algebraic value, the instrument is broken and no number
477 from the run may be reported.

478 The first cause was that the candidate populations were alpha-blends of two different cell lines, so
479 their two subpopulations sit in two different contexts. An additive predictor adds one delta *per context*, so
480 the quantity measured was $\cos(\delta_{A549}, \delta_{K562})$: the similarity of two contexts’ deltas, which has no reason
481 to be 1 and is not a test of additivity. The theorem is a *within-context* statement, and only a within-context
482 experiment can test it. We therefore re-ran it with both subpopulations drawn from one cell line’s own
483 control cells and the predictor run separately on each, so the partition is true by construction (Methods).

484 **The second cause is a result in its own right, and it sharpens the argument rather than weaken-**
485 **ing it.** With the construct fixed, the gene-space additive predictor returns $\cos = 1.000000$ exactly, as
486 required. The *latent* additive predictor does not. It returns values as low as -0.13 , and the reason is that
487 a latent model does not add a delta to a cell: it *reconstructs* the cell and adds a delta to the reconstruction,
488 returning $P(C) + W\delta_z$ where P is the autoencoder’s reconstruction. Referred to the model’s own de-
489 coded baseline the reconstruction term cancels, and the induced cosine is exactly 1.000000: the theorem
490 holds and the model is additive, as claimed. Referred to the *real* control cells, which is what a retrieval
491 scorer necessarily does, the response carries $\text{mean}(P(C_k)) - \text{mean}(C_k)$, the reconstruction bias of sub-
492 population k , which is cell-state dependent, differs between subpopulations (their biases are themselves
493 near-orthogonal, $\cos = -0.05$ to -0.18), and does not cancel.

494 **So an additive latent predictor induces exactly zero real response divergence and yet, to any scorer**
495 **that measures it, appears to induce a great deal.** That apparent divergence is the generator’s reconstruc-
496 tion error. A distributional score run on such a candidate population is not merely blind to subpopu-
497 lation structure that is not there; it is confidently scoring an autoencoder’s artefacts and returning a
498 number about nothing. This is the same lesson as the rest of the paper, reached a third way: what you

measure against decides what you find. It also means that a structure-preservation or divergence diagnostic evaluated on a latent model’s output against real controls measures the autoencoder, not the perturbation, and that includes the diagnostics we report above.

Gate 2, recoverability. Even where differential structure exists, a decision layer must be able to resolve it. We test this on a controlled mixture with ground-truth labels (K562 at 10 μ M, HDAC-class versus JAK-class cells), where the bimodal structure exists by construction. The question that matters is not whether clustering succeeds, but *why it fails*, and those are two very different papers. If the information is absent from the representation, no method can help. If it is present and only the unsupervised assignment step cannot reach it, the bottleneck is an algorithm and is solvable. Distinguishing them requires an upper bound, which prior work in this area, ours included, has not reported.

We therefore give every method the same representation and score them all in the same unit, best-permutation accuracy against the true labels, granting each clustering its label matching for free. Unsupervised recovery is poor: *k*-means 0.632, Gaussian mixtures 0.661, Leiden 0.674, and HDBSCAN 0.500, which finds no density structure at all and assigns every cell to noise (Fig. 6e). The first two are handed the true *k*; the last two are the graph- and density-based methods whose absence was the obvious objection to our earlier analysis, and adding them does not change the answer.

The supervised ceiling is the informative number. Handing a classifier the ground-truth partition and fitting the representation inside the training fold, held-out accuracy reaches only **0.692** (AUC 0.752), and a nonlinear learner does not beat a linear one (random forest 0.669, *k*-nearest-neighbours 0.605, logistic regression 0.685). The boundary is not a hidden nonlinear one that our probe was too weak to find. It is barely there.

The gap between the ceiling and the best unsupervised method is +0.018 accuracy. Unsupervised clustering is already extracting essentially everything this representation contains about this partition, so *within this benchmark* the limit is informational rather than algorithmic. Two later results retract almost everything one might conclude from that sentence, and we flag them here rather than let a reader carry it forward. The gap is an *underestimate*: the design gives the clusterers the easier transductive representation and the probe the harder in-fold one, which shrinks the gap and so flatters this very conclusion (we had stated that bias backwards; Methods). And the 0.692 ceiling is itself largely an artefact of *pooling* several HDAC and several JAK compounds into two classes: split the same cells one drug against one drug and the ceiling rises to 0.879 (Fig. 5d). The informational reading does not survive either correction, and it does not survive contact with tissue.

The probe is not simply insensitive: raise the separation between the two states artificially and it pulls away from clustering exactly as it should (ceiling 0.828 versus 0.714 at $1.5\times$ separation, 0.924 versus 0.762 at $2\times$; Fig. 6f), so its failure to pull away here is a measurement rather than a null result. Consistent with it, across the 529 (cell line, drug) pairs in SciPlex3 with enough cells to cluster (188 drugs $\times$ three lines, less those too small; the unit is the pair, not the drug) the median silhouette of a two-way split is 0.034, with only 0.4% of drugs exceeding 0.1, and the earlier nine-column clustering sweep this analysis supersedes is reported in Extended Data Fig. 3.

But this benchmark is one we built, and the next section shows that it misled us. An HDAC-treated and a JAK-treated K562 cell are a mixture we assembled; they are not a subpopulation structure that occurs in a patient. Read on its own, the number above would license the conclusion that real

subpopulations sit at an information ceiling no method can exceed, and we drew exactly that conclusion before testing it. It is wrong.

What Gate 2 does and does not constrain. Energy distance, MMD and sliced-Wasserstein are *cluster-free*: they compare whole populations and never form a partition, so Gate 2 cannot be read as a claim that they require one. The constraint it imposes is subtler and it closes the loop with the functional-oracle result. A distributional distance is a single number into which every source of population difference is summed. Where the subpopulation signal sits at an information ceiling as low as the one measured here, that signal cannot be what dominates the distance; something else must, and we have measured what: the candidate’s own response magnitude, which tracks the energy distance at $\rho = +0.791$. This is why the distributional score behaves like a magnitude scalar on the independent oracle. Gate 2 does not stop a cluster-free score from being computed. It determines whether the number that score returns is about subpopulation structure or about something else entirely.

The two gates therefore compound, and their consequences differ by score type. A distributional decision layer cannot exploit subpopulations that do not respond differently (Gate 1), and where they do respond differently it cannot rely on telling them apart (Gate 2). Note what Gate 1 is *not*: it is not structure preservation. Every predictor we tested preserves structure and every one still fails, so a generator validated on a structure-preservation metric would pass a test that does not bear on the question. Coverage-type scores, which need the partition explicitly, are blocked by Gate 2 *wherever it is shut*, which in this study is the constructed cell-line mixtures and not the patient tumours of the next section. Global scores, which do not need a partition, are never blocked but are left measuring whatever channel is largest, which is not the one their advocates intend.

Two further tests, both consistent with the criterion and neither rescuing it, are given in Supplementary Note 2: coverage performance rises with query identifiability but only weakly and not monotonically, and when queries are stratified by the Gate-2 variable itself the distributional advantage is flat.

HIR-Bench separates observable from circular predictors

To study these regimes under controlled conditions we construct HIR-Bench (Heterogeneous Inverse Retrieval Benchmark), a synthetic, method-neutral generator with a specified latent welfare oracle (Fig. 5). Each task instantiates one of three information conditions, observed, predicted_mean (candidates collapsed to their means), and predicted_structure (candidates with partially preserved structure), and the generator admits an analytical flip boundary $\alpha^* = B / (A + B)$ separating the regime where distributional retrieval can help from where it cannot (Fig. 5a).

The grid comprises 13,440 benchmark *instances* (672 parameter configurations $\times$ 20 seeds; these are synthetic retrieval instances, not biological cells). The sanity checks hold (no-conflict flip rate 0.0000; median boundary margin 0.27; Extended Data Fig. 6d), and collapsing candidates to their means inflates decision regret most severely against the worst-welfare objective, the one that depends on the minority subpopulation (0.29 observed versus 0.96 predicted-mean; Extended Data Fig. 6c).

Is retrieval failure predictable, and from what? The second half of that question matters more than the first, and a benchmark with a latent oracle is the only place it can be answered cleanly. Features derived from the oracle are available to the benchmark’s authors but never to a retrieval method at query time; features of the observed populations are available to both. Fitting the same model on each feature set separates a usable diagnostic from a circular one, and the two answers are opposite.

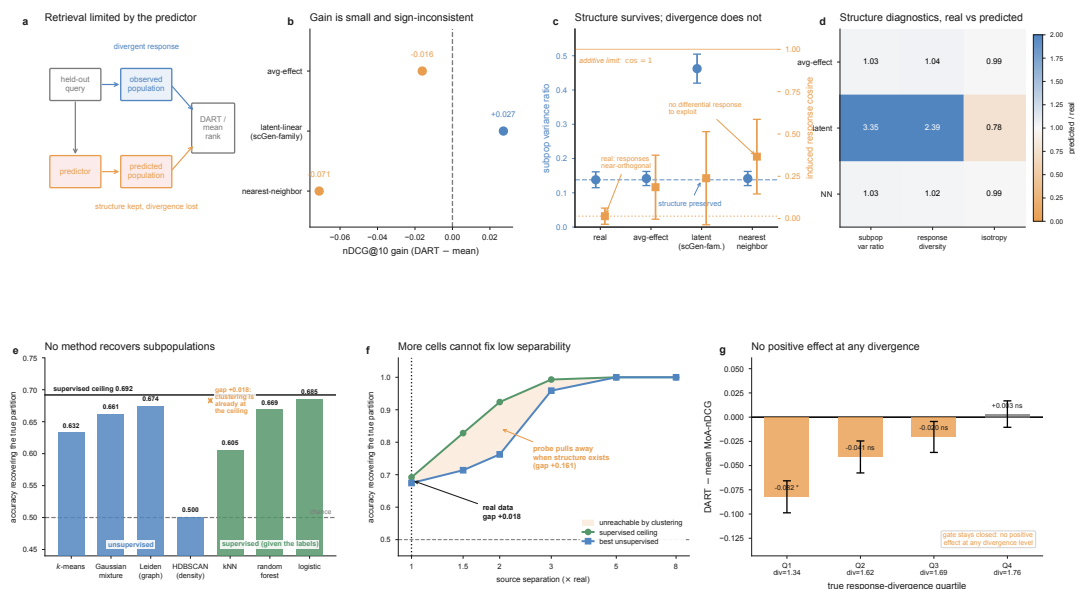


Figure 6: The information condition is a two-gate criterion: candidate populations must show differential response, and that structure must be identifiable. **a**, The predict-then-rank design: a held-out query is scored against candidate populations that are either observed or produced by a perturbation predictor, and the two gates are the two properties such a candidate population must have. **b**, Predictor-specific retrieval gain (nDCG@10, DART minus mean). The gain is small and *not* consistent in sign: -0.071 (nearest-neighbour), -0.016 (average-effect), $+0.027$ (latent-linear). **c**, Gate 1. Predicted candidate populations *retain* baseline subpopulation structure (variance ratio 0.142 versus 0.138 for real ones, left axis) but do not reproduce *divergence*: the induced response cosine between the two subpopulations is 0.19–0.37 for predicted populations against 0.014 for real ones (right axis). An additive predictor adds one delta vector to every cell of a context, so within that context its two subpopulations have identical responses: cosine exactly 1, divergence exactly zero. That algebraic ceiling is marked on the right axis. **The predicted values shown here do not measure it, and we do not interpret them.** These candidate populations are blends of two cell lines, so their subpopulations sit in different contexts and an additive predictor emits a different delta for each; and for a latent model the response taken against real control cells also carries the generator’s reconstruction bias. Measured within one context and against the model’s own baseline, an additive predictor returns exactly 1.000000, as the algebra requires (Methods). **d**, Structure diagnostics, predicted relative to real. **e**, Gate 2, on a known bimodal mixture (K562, HDAC-class versus JAK-class cells). All methods are scored in one unit, best-permutation accuracy against the true labels, with the label matching given to each clusterer for free. Unsupervised recovery is poor (*k*-means 0.632 and Gaussian mixtures 0.661, both handed the true *k*; Leiden 0.674; HDBSCAN 0.500, finding no density structure at all), but the *supervised ceiling*, a classifier given the ground-truth partition with its representation fit inside the training fold, reaches only 0.692 (AUC 0.752), and a nonlinear learner does not beat a linear one (forest 0.669, *k*NN 0.605, logistic 0.685). The gap between the ceiling and the best unsupervised method is **0.018**: clustering is already extracting essentially everything this representation holds, so *within this benchmark* the limit is informational rather than algorithmic. That qualifier is load-bearing, and two later corrections dismantle what it appears to license. The gap is an *underestimate*, because the clusterers are given the easier transductive representation and the probe the harder in-fold one, which shrinks it (Methods). And this 0.692 ceiling is largely an artefact of *pooling* several HDAC and several JAK compounds into two classes: the same cells split one drug against one drug give 0.879, and in a patient’s tumour the same drug-response question gives a ceiling of 0.923 with unsupervised clustering reaching only 0.777 (Fig. 5d). In real tissue the limit is algorithmic, not informational, and the reading offered here is withdrawn. **f**, The control that licenses reading panel **e** at all. Raise the separation between the two states artificially and the probe pulls away from clustering exactly as it should (ceiling 0.828 versus 0.714 at $1.5\times$; 0.924 versus 0.762 at $2\times$), so its failure to pull away on this mixture (leftmost, separation = 1) is a measurement rather than an insensitive probe. **g**, The conditional advantage a working information condition would produce is absent. DART-minus-mean MoA-nDCG gain by *true* response-divergence quartile (bars are means; Benjamini–Hochberg corrected). No stratum is positive. The lowest-divergence quartile is significantly *negative* (mean -0.082 , median -0.030 , $q = 5.8 \times 10^{-5}$, $n = 166$); the rest have a median of exactly zero and means of -0.041 , -0.020 and $+0.003$, none significant. Where the theory predicts the largest gain there is none. Stratifying instead by the Gate-2 variable leaves the advantage flat (Supplementary Note 2).

We therefore run a 2×2 design: two feature sets crossed with two cross-validation groupings. The *observable* features (subpopulation-variance ratio, isotropy and response diversity of the query and candidate populations) are computable at query time. The *oracle-derived* features (top- k disagreement, Kendall conflict) are each a function of the benchmark’s latent utility matrix, as is the label. The groupings differ in their unit: the label is a deterministic step function of $(\alpha, \text{conflict})$, so the 13,440 instances contain only 28 independent parameter cells, whereas the natural leave-one-instance-out scheme treats 672 as independent and leaves roughly twelve near-identical copies of each held-out instance in training.

All four cells are shown in Fig. 5b. Read across the top row, retrieval failure is moderately predictable from what a method can see: AUC **0.788** against a majority-class rate of 0.643, the query’s response diversity carrying the highest weight. Read across the bottom row, the oracle-derived features are no better than chance (0.400; with 28 independent cells this is not distinguishable from 0.5). They predict one function of the latent utility matrix from another, which measures circularity, not predictability.

The two columns separate the two failure modes cleanly, and the interaction is the point. Pseudo-replication inflates the circular feature set by +0.240 (0.400 to 0.640) but the honest feature set by only +0.048 (0.788 to 0.836). An instance-level holdout preferentially rescues features that carry no generalizing signal, because memorizing near-duplicates is all they have; features with real signal barely benefit. A benchmark whose cross-validation unit is finer than its label-determining axes will therefore manufacture apparent predictability precisely where there is none, and the effect is largest for exactly the features an author is most tempted to use. Neither effect is specific to this benchmark.

Two caveats bound the 0.788. The effective sample size is 28 parameter cells, so a confidence interval computed on $n = 13,440$ would be meaningless, and by the same token the gap between 0.400 and 0.5 is not itself interpretable. And the label is a deterministic step function of those parameters, which makes every held-out fold single-class, so no per-fold AUC distribution exists and the pooled out-of-fold AUC is the only statistic available. “Moderately predictable” is as strong a claim as this supports.

Two limitations are essential. HIR-Bench is intentionally constructive: built from a specified latent welfare model, its phase boundary is expected by design, and the question that matters is whether real data project onto the same regimes. They do not: the boundary fails to transfer, scoring 61.1% against a 95.4% majority-class baseline (Methods). And one apparently supportive number is not evidence: the gate “correctly” flags mean-collapsed predicted responses as no-DART in 100% of the transfer test, but that rate is a tautology, since the rule returns no-DART unconditionally for that information condition (Supplementary Note 1). HIR-Bench is a controlled framework for exposing failure modes, not a real-world diagnostic.

In real tissue Gate 2 is an algorithmic limit

Every real-data positive in this study so far lives in cell-line mixtures that *we* constructed. A minority subpopulation assembled by pipetting A549 together with K562 is not evidence about the heterogeneity a tumour actually has, and a criterion validated only on such a mixture is a criterion validated on our own pipetting. We therefore tested the two gates on ZhaoSims2021, acute-slice culture and surgical biopsy from ten glioblastoma patients, in which malignant glioma cells, tumour-associated myeloid cells and oligodendrocytes co-exist inside one patient’s tumour. Nobody mixed anything. Compartments are assigned by validated marker score and cells without a confident call are dropped rather than reassigned; an earlier attempt that took a bare argmax over five marker sets fabricated a 22% “T-cell” compartment

622 whose cells did not express CD3 (Methods). This is *not* a retrieval benchmark and we do not use it as
623 one: the panel is six drugs and only one patient received more than two.

624 **Gate 2 must be measured on the same construct in both settings, and our first attempt was not**
625 (Fig. 5d). The constructed Gate-2 benchmark separates HDAC-treated from JAK-treated K562 cells: one
626 lineage, cells differing only in which drug they saw, a *drug-response* partition. Our first natural test sep-
627 arated malignant from myeloid *control* cells: two lineages, a *cell-type* partition, and one that is trivially
628 easy because a glioma cell and a macrophage differ in thousands of genes. It reached a ceiling of 0.964,
629 and comparing that to 0.692 compared two different questions. Gate 2 is defined on the partition a
630 retrieval score must resolve inside a candidate population, which is a drug-response partition. We there-
631 fore re-ran it as one: cells treated with drug A against cells treated with drug B, *within one compartment*,
632 *within one patient*, which is the K562 construct transplanted into tissue nobody assembled (36 drug-pair
633 splits; Methods). We additionally report the ceiling under the representation the clusterers actually re-
634 ceive, because the in-fold representation used earlier deflates the ceiling and so *shrinks* the very gap the
635 conclusion rests on.

636 **Two things follow, and the second one reverses us.** First, the constructed ceiling of 0.692 was sub-
637 stantially an artefact of *pooling drugs within a class*. Splitting the same K562 cells one drug against one
638 drug, as the natural test necessarily does, lifts the ceiling to 0.879 with unsupervised clustering at 0.837.
639 Much of what we read as an information limit was the within-class heterogeneity of several distinct
640 compounds we had merged.

641 Second, and this is the finding: in a real tumour the information is plainly *present*, and unsupervised
642 clustering *cannot reach it*. The supervised ceiling is 0.923 while the best unsupervised method reaches
643 only 0.777, a gap of +0.117 against +0.007 on the like-for-like constructed split. Centroid-based meth-
644 ods collapse (*k*-means 0.638, Gaussian mixtures 0.620) where the graph-based method partly survives
645 (Leiden 0.780), which is the signature of a dominant variance direction that is not the drug-response
646 axis: inside a real compartment, cell-state heterogeneity outweighs the drug response and unsupervised
647 partitioning locks onto it.

648 **We therefore withdraw the information-limit claim, and we withdraw the first withdrawal too.**
649 The original conclusion, that real subpopulations sit at an information ceiling no method can exceed, is
650 wrong. But so is the reading we replaced it with, that real subpopulations are simply easy to resolve;
651 that came from the cell-type comparison and does not survive being posed as a drug-response question.
652 In real tissue Gate 2 is an *algorithmic* bottleneck, not an informational one: the partition is there to be
653 found, off-the-shelf clustering does not find it, and the gap is therefore something a better method could
654 close. That is the one place in this study where the news for distributional retrieval is constructive rather
655 than cautionary, and it is the opposite of what we concluded from our own benchmark twice.

656 **We attacked that gap three ways before believing it, because it is the only constructive claim in**
657 **this paper and it was the least defended number in it.** First, it is a difference of two *maxima* (best of
658 three probes minus best of four clusterers), and a maximum over methods is upward-biased. Recomput-
659 ing every fixed (probe, clusterer) pair, so that no selection occurs anywhere, gives +0.122 in the tumour
660 against -0.002 on the constructed split: the same conclusion, and the maximisation moves the natural
661 number by only -0.005. (Pairs involving HDBSCAN are excluded from that figure and reported sepa-
662 rately: it assigns every cell to noise and scores exactly 0.500, so a large “gap” against it means only that

the clusterer did nothing.) Second, the 36 splits are not 36 independent observations, since 30 come from one patient and reuse the same six drugs; a bootstrap over splits would be the pseudo-replication this paper criticises, so we resample *patients*. The gap is +0.117 with a 95% interval of [+0.115, +0.135], and the natural-minus-constructed difference is +0.109 [+0.052, +0.129], with none of 5,000 bootstrap draws at or below zero. Third, dropping PW030 altogether leaves six splits in three patients, and the gap there is +0.123 [+0.098, +0.138]: slightly *larger*, not smaller.

Three limits nonetheless bound it. A cluster bootstrap with only *four* clusters is known to under-cover, so the interval above should be read as a floor on the true uncertainty rather than a faithful estimate of it. *Cells were dropped:* 39.9% fail the compartment-assignment margin and are excluded rather than reassigned (Methods, Supplementary Table 3), and because the discarded cells are by construction the hardest to place, both the ceiling and the clustering accuracy are optimistic. *The drug panels differ:* the constructed split contrasts HDAC with JAK inhibitors and the natural one six unrelated compounds, so response magnitude is not matched across the arms. We report the comparison because it is the closest to like-for-like these data permit, and we claim no more precision for it than four patients and one drug panel can support.

Gate 1 opens too, and the same benchmark misled us in the opposite direction (Fig. 5d). Naturally co-existing compartments do respond in different directions to the same drug, but far less so than our mixtures implied: the induced response cosine has a median of 0.566 across 17 patient-drug pairs, where our constructed mixtures sit at 0.014–0.044, essentially orthogonal. One drug pushes two unrelated cell lines in unrelated directions; malignant cells and tumour-associated macrophages in one patient share most of their response. The constructed benchmark overstates the divergence a distributional score can exploit, by roughly twofold on a divergence scale ($1 - \cos$: 0.99 against 0.43), while misplacing the identifiability limit. (The ratio of the cosines themselves, 40-fold, is not a meaningful measure of divergence when one cosine is near zero.) Both distortions flatter this paper’s premise, and neither was intentional.

Both gates are therefore open in real tissue, and the two-gate account predicts that a distributional advantage should follow. The mechanism that would produce one is missing. The paper’s motivating scenario, that two drugs can share a mean signature and do opposite things to a subpopulation, can be tested directly on a real tumour. Across 18 within-patient drug pairs, the similarity of two drugs’ *mean* signatures ranks the similarity of their *malignant-compartment* responses at Spearman $\rho = +0.878$ (Fig. 5e). Where the mean says two drugs are alike, the malignant compartment mostly agrees. Of the five drug pairs most similar on the mean, two have a malignant response meaningfully less similar than their mean suggests, so the residual is real; it is also small.

Two limits on that number, and they are severe. First, *it is close to a single-patient result*: 15 of the 18 pairs come from one patient (PW030, the only one given more than two compounds) and reuse the same six drugs, and the remaining three patients contribute one pair each. A *p*-value computed as though the 18 pairs were independent is anticonservative by a wide margin, and we therefore do not quote one; the honest description is a strong association in one patient, consistent in sign with three single-pair observations in three others. Testing the premise properly needs a cohort in which many patients receive many drugs, which this dataset is not. Second, a rank correlation between two similarity scores is not a fraction of variance and not a statement that the response vectors are collinear, and we do not read it as

704 either.

705 **And we did not run retrieval here, so we do not report a retrieval result.** ZhaoSims2021 supports
706 six compounds with only one patient receiving more than two; a leave-one-drug-out Hit@1 computed
707 on that would be an anecdote wearing a benchmark’s clothes. What this section shows is that the *mech-*
708 *anism* a distributional advantage would have to run through, mean-similar drugs diverging inside a
709 subpopulation, is largely absent in real tissue. That is an inference about why an advantage should not
710 be expected, not a measurement of its absence.

711 **The two gates are necessary but not sufficient, and this is how we found out.** Opening them does
712 not deliver a distributional advantage, because in real tissue the compartment-specific response is mostly
713 ranked by the mean already. The independent functional oracle reaches a compatible conclusion by an
714 entirely different route: energy retrieval beats the mean incumbent there, and a scalar magnitude control
715 then reproduces almost the whole of that margin. The two lines of evidence share no data, no metric and
716 no code path, and they do not share a number either, so we do not claim they converge on one. What
717 they agree on is a direction: distributional information in this domain is real, positive, and much smaller
718 than objective-aligned evaluation makes it appear.

719 Discussion

720 The main result of this work is not that DART is better or worse, but that *how much* better it is depends
721 almost entirely on who is asked. Under a metric of its own distributional kind the advantage is large.
722 Under an independent, semantically matched functional oracle it is real but small, and almost all of what
723 remains is a response-magnitude channel that a scalar reproduces without comparing any distributions
724 at all. Under a task-proximal mechanism criterion it is absent. The same rankings, the same scorer, the
725 same queries: only the judge changes. DART is valuable here as a probe, an instrument that can be scored
726 by its own objective and by more independent criteria on the same tasks, rather than as a therapeutic
727 recommender. The paper’s positive anchors survive intact: the unification of mean and distributional
728 retrieval is an algebraic fact, and there is genuine distributional signal in response-available populations.
729 What does not survive is the size of the gain that objective-aligned evaluation reports for it.

730 We also asked, as a direction for future work rather than a load-bearing result, whether the re-
731 sponse divergence a distributional score measures has any biological correlate. In the Frangieh Perturb-
732 CITE-seq data, cells whose response direction diverges most between contexts are weakly enriched for
733 resistance-associated programs. Without a drug timecourse, a survival readout or a post-treatment clone
734 measurement this is hypothesis-generating only, we make no claim that DART predicts resistance, and
735 the analysis is reported in Supplementary Note 3 and Extended Data Fig. 4 rather than here.

736 Why the negative results are central

737 The predict-then-rank nulls and the independent-metric results are the evidence that supports this pa-
738 per’s core claim, not a shortfall against it. The objective-aligned positive on its own would license the
739 conclusion that distributional retrieval substantially improves ranking. The task-proximal metrics do
740 not support that, and the independent functional oracle supports a far weaker version of it: the distri-
741 butional score does beat its incumbent there, but a scalar that performs no retrieval reproduces nearly
742 the whole margin. The positive and the null together, labeled by metric class, are what tell a practitioner
743 which conclusions a given metric licenses and which it does not.

744 Implications for virtual-cell and perturbation-prediction models

745 The two-gate result has a direct implication for the fast-growing class of perturbation predictors and
746 virtual-cell foundation models. In this framework such models are candidate-response sources, and
747 whether a distributional decision layer can add anything on top of them depends entirely on whether
748 they supply *differential* response. The load-bearing statement here is algebraic and needs no benchmark
749 to establish: a predictor that adds one delta vector to every cell induces exactly zero response divergence,
750 so additivity is *sufficient* to make a distributional decision layer worthless on top of it. That is a property
751 of a model class, not a deficiency of any implementation, and no amount of training removes it.

752 What we do not claim is that today’s predictors are, as a field, additive. We can say it of one deployed
753 model: the published scGen variational autoencoder passes its learning check and sits at a true induced
754 divergence of 0.972 against the additive ceiling of 1.000, reproducing about 3% of the divergence real
755 subpopulations show. That is a fact about scGen, not about the field. The in-repository linear-latent
756 model is additive by construction, so its value demonstrates the algebra rather than surveying anything.

757 Non-additive, cell-state-conditional predictors exist, including optimal-transport, flow and
758 Schrödinger-bridge formulations^{16,17}, and a latent-space optimal-transport map does open Gate 1,
759 though only a crack. What we have not done is the second half of the test. A predictor that genuinely
760 produces state-specific responses should, if the two-gate framework is right, restore a distributional
761 advantage that additive predictors cannot, and one that opens Gate 1 yet yields no such advantage
762 would falsify the framework. That test is out of reach for the predictors we could run: the optimal-
763 transport map’s cross-line predictions are too weak to retrieve with, and the flow model this test most
764 plausibly calls for, CellFlow²², learns and opens Gate 1 within a context but does not carry that learning
765 to a held-out cell line, because per drug the transfer is bounded by the drug’s cross-context response
766 similarity and the retrieval task selects exactly the drugs for which that similarity is near zero (Results).
767 The decisive test therefore still requires a predictor that both predicts accurately *across contexts* and
768 opens Gate 1, most plausibly a flow or Schrödinger-bridge model trained to transfer; we name this as
769 the test that would falsify us, and record that the flow model we trained does not yet complete it.

770 The methodological point stands independently of that test, and it is sharper than we first realised.
771 Differential response, not mean-level accuracy and not even structure preservation, is the property that
772 determines whether distributional retrieval is applicable at all. Structure-preservation metrics alone
773 would have scored every predictor we tested as adequate. Worse, they would have scored the gener-
774 ator’s own reconstruction error as biological structure. An additive latent model induces exactly zero
775 true response divergence, yet measured against real control cells, as a scorer must, it appears to in-
776 duce a great deal, because the autoencoder reconstructs the two subpopulations with different biases.
777 A distributional decision layer stacked on such a model would be scoring the autoencoder. Anyone
778 reporting that a generative perturbation model “preserves subpopulation structure” or “induces diver-
779 gence” should therefore state which baseline the claim is measured against, and show that it survives
780 being measured against the model’s own output. This is consistent with independent benchmarking,
781 in which foundation models and deep-learning predictors did not outperform deliberately simple base-
782 lines for perturbation prediction^{11,23}: objective-aligned reconstruction accuracy need not translate into
783 an advantage over mean or linear baselines.

784 What independent validation would require

785 The external functional readout we could reach, GDSC dose-response profiles matched to our drug
786 panel, constrains the question without settling it. It is a semi-real oracle: viability is measured in a
787 separate bulk assay, matched by compound name, under a different dose regime. It establishes that
788 transcriptional similarity recovers functional similarity better than a mean signature does, and that al-
789 most all of that recovery is a magnitude channel. It cannot establish therapeutic utility. Turning DART
790 from an evaluation probe into a therapeutic method would require readouts measured on the same cells
791 the method is asked to treat: dose-response viability *in situ*, longitudinal treatment survival, functional
792 perturbation validation, post-treatment clone expansion, and combination-response validation. Until
793 then, the defensible claim is that distributional information is *present* and *modestly useful*, and far smaller
794 than objective-aligned evaluation makes it appear.

795 We urge four specific controls on anyone reporting a validation of a distributional retrieval method,
796 each of which caught an error of ours.

797 First, match the oracle to the task. A similarity retriever graded on absolute potency is being asked
798 a question it was never posed, and it will fail in a way that looks like a finding: potency tracks response
799 magnitude, a distributional distance also tracks response magnitude, and the two confound. We ran that
800 comparison, drew a conclusion from it, and had to withdraw the conclusion; it now appears only as an
801 audit.

802 Second, beat the magnitude scalar. Rank the candidates by how closely their response magnitude
803 matches the query's, compare no distributions at all, and show that your method beats it. On the
804 matched oracle ours does not do so significantly, and on the mismatched one it loses outright. A dis-
805 tributional method that cannot clear this bar has not shown that the distribution is what is doing the
806 work.

807 Third, vary the shape of your external oracle, not just its identity. It is natural to assume that once
808 a criterion is external, independent and invisible to the method, it can be trusted to arbitrate. It cannot.
809 Two oracles built from the same proteins in the same cells, one computed as a mean and one as a distri-
810 bution, hand victory to opposite methods (Fig. 5c). A field that validates distributional methods against
811 distributional readouts and mean-based methods against mean readouts will find both vindicated, and
812 neither finding will be about biology. The practical form of this control is cheap: compute your external
813 criterion in at least two statistical forms and report both. If your method only wins under the form that
814 matches its own, say so.

815 Fourth, and least comfortable, do not let a benchmark you built decide a claim about biology, and
816 check that the two things you are comparing are the same question. Our constructed cell-line mixtures si-
817 multaneously understated how identifiable real subpopulations are (ceiling 0.879 on a like-for-like split,
818 against 0.923 in a patient's tumour, where our class-pooled mixture had said 0.692) and overstated how
819 divergently they respond (cosine 0.014 against 0.566). Both distortions happened to favour the premise
820 we were testing, and neither was visible from inside the benchmark. Worse, our first attempt to check
821 the benchmark against tissue compared a drug-response partition in the mixture with a *cell-type* par-
822 tition in the tumour, which is a far easier problem, and it produced a reassuring number (0.964) that
823 meant nothing. A comparison across two settings is only as good as the sameness of the question asked
824 in each, and it is alarmingly easy to hold the protocol fixed while silently changing the question. Any

825 information-condition criterion validated solely on synthetic or constructed heterogeneity should be
826 treated as a statement about that construction until it is checked against tissue, and the check must ask
827 tissue the same question.

828 **Field-level evaluation implications**

829 The broader lesson is that the central distinction is not DART versus mean retrieval, but objective fidelity
830 versus independent utility. Our field-level audit shows that objective-aligned evaluation is the default
831 across signature retrieval, expression prediction, and graph-based inverse design: the headline metric is
832 usually a restatement of the method’s own objective. This is legitimate for testing objective fidelity, and
833 we do not claim any prior method is wrong. But it means that strong benchmark numbers, on their own,
834 answer a narrower question than they are often read to answer.

835 Two corollaries follow. A benchmark whose cross-validation grouping is finer than its label-
836 determining axes will manufacture apparent predictability from nothing, as ours does for oracle-derived
837 features; and a distributional score’s apparent link to therapeutic value can be a response-magnitude
838 confound, so any such claim should be required to beat a query-independent magnitude baseline.
839 Neither failure mode is specific to this work. This reading is corroborated by recent metric-focused
840 studies: the distributional metrics used here have documented failure modes, with the Wasserstein
841 distance degrading under variance scaling in high-dimensional expression space and the energy
842 distance overlooking disruptions in gene–gene dependencies²⁰, and metric choice alone can flip
843 model rankings²¹. We cite the first against ourselves; it bounds the interpretation of our own Class-A
844 scores. The contribution is to make the objective-fidelity/independent-utility distinction explicit and
845 measurable on a single task set, a template other inverse-retrieval methods could adopt.

What this study does not claim

1. **We do not claim that the predictors tested here represent the field.** The in-repository linear-latent model (PCA encoder, additive latent delta, linear decoder) is additive *by construction*, so its Gate-1 value demonstrates an algebraic property rather than surveying deployed models. We *do* now report one deployed model: the published scGen variational autoencoder passes its learning check and returns a true induced divergence of 0.972 against an additive ceiling of 1.000, so it is near-additive in practice. That is one model, not a field.
2. **We show that a non-additive predictor opens Gate 1, but not that opening it buys retrieval gain, and the second half is the experiment that would falsify us.** A latent-space entropic optimal-transport map passes its learning check (0.352) and reaches a true induced divergence of 0.900, significantly below the additive ceiling of 1, so Gate 1 does open; but only a crack, since real cells sit at 0.205, and the map's learning is weak. The same map in gene space did not clear the bar, and the published CPA model, though it reconstructs cells at $R^2 \approx 0.39$, does not learn per-drug effects at either sixty or three hundred cells per drug (delta learning 0.08 in both) and is therefore void. We did re-run the predict-then-rank retrieval with this Gate-1-opening map, and it was inconclusive: the OT map's cross-line predictions are too poor to retrieve with at all (Hit@1 0.05 under every rule, against 0.88 for the additive predictor), so we lack a predictor that both predicts accurately and opens Gate 1, and whether opening Gate 1 buys a distributional advantage stays untested. A modern flow-matching model, CellFlow²², conditioned on ECFP4 compound fingerprints, learns within context (mean-delta cosine 0.24, opening Gate 1 at induced divergence 0.85) but collapses to 0.03 under leave-one-context-out at a matched budget; per drug this transfer tracks cross-context response similarity ($r = 0.85$), so it too cannot serve as the clean cross-context predictor the falsification requires. The collapse holds under distributional (energy) and differential-expression metrics, not only the mean-delta check; and conditioning CPA on the same ECFP fingerprints instead of a learned embedding does not rescue it (learning 0.03), locating that failure in the adversarial objective rather than the drug representation.
3. **We do not claim that DART provides better therapeutic recommendations.** Every gain in this study is measured under an objective-aligned (Class A), task-proximal (Class B), or semi-real functional (Class C) criterion. None is a clinical or same-cell endpoint.
4. **We do not claim that the functional-oracle advantage is a distributional effect, nor that it is robust to how the oracle is built.** Energy retrieval does beat the correctly specified mean incumbent under the cell-line-centered GDSC oracle (+0.276 versus +0.083), but a query-dependent scalar that compares no distributions reaches +0.232, energy beats it on only 62 of 103 queries ($p = 0.073$, itself anticonservative), and partialling that magnitude channel out leaves +0.097: the gain is real and mostly not distributional. It is also conditional on the centering, since under the uncorrected oracle energy falls to +0.097 and loses to both the mean incumbent (+0.138) and the scalar (Supplementary Table 4). We specified the centering in advance and argue it is correct, but only the negative survives both constructions: under neither oracle does the distributional score significantly beat a scalar that compares no distributions.
5. **We do not claim that retrieval similarity is anti-aligned with therapeutic utility.** An earlier framing of this work graded the rankings against *absolute potency* and drew that conclusion. It does not

848 Methods

849 Data sources and preprocessing

850 Three single-cell datasets carry every result reported here. SciPlex3²⁴: 276,325 cells $\times$ 2,000 genes across
851 three contexts (A549, K562, MCF7) and 188 drugs, all present in all three contexts. Frangieh melanoma
852 Perturb-CITE-seq²⁵: 218,023 cells $\times$ 2,000 highly variable genes across Control, IFN γ , and Co-culture
853 conditions with 239 CRISPR knockouts. CD34+ (GSE306429): 33,984 cells $\times$ 2,000 genes, 36 drugs, used
854 as an honest negative control. A fourth resource, the LINCS closed-loop / PDGrapher benchmark (41,070
855 rows across A549, MCF7 and PC3, 1,369 drugs), was prepared for a graph-based inverse-design compar-
856 ison that we do not report; *no number in this paper is derived from it*, and we name it only because it ships
857 with the code. A fifth, GDSC dose-response, is the external potency oracle used for the Class-C analysis
858 and is described below. Expression is processed to log-normalized differential profiles against matched
859 controls; per-context control means and treated-population rows are exposed through a common dataset
860 interface. Dataset scale, preprocessing settings and query construction are tabulated in Supplementary
861 Table 1.

862 DART retrieval task and distributional metrics

863 Given a target population Q and candidate populations $\{P_d\}$, DART ranks by a population-to-
864 population distance. Implemented scores: the energy distance (subsampling to a fixed maximum
865 cell count with a fixed seed), an RBF maximum-mean-discrepancy, a sliced-Wasserstein distance (128
866 projections), and a subpopulation-coverage aggregate with a temperature parameter β interpolating
867 from mean aggregation ($\beta \rightarrow 0$) to worst-case ($\beta \rightarrow \infty$). Mean-signature retrieval, by cosine or by
868 Euclidean distance between mean signatures, is the zero-variance limit of the same family.

869 Signature and PCA baselines

870 CMap-style signature retrieval is implemented with two selectable scoring rules. The cosine rule com-
871 puts the cosine of mean-delta signatures and is the same operation as mean-cosine retrieval, which is
872 the source of the exact Hit@1 identity. The WTCS rule is a rank-based (GSEA-style) weighted connectiv-
873 ity score, a monotone sibling that is not algebraically identical. PCA-latent retrieval fits PCA per query
874 and scores in latent space by mean or distributional distance.

875 Predict-then-rank baselines

876 Candidate responses are generated by an average-effect predictor (per-drug mean deltas), a nearest-
877 neighbour predictor (nearest training context by control-state similarity), and a CPA-style linear-latent
878 predictor, used as a surrogate for the latent-arithmetic family that scGen⁵ belongs to; we do not bench-
879 mark a trained scGen, and the active backend is recorded in every output. They are used purely as
880 candidate-response *sources*, not as models under benchmark. They do not stand in for the broader class
881 of published perturbation predictors and virtual-cell foundation models (scGen⁵, CPA⁶, chemCPA¹³,
882 GEARS⁷, scGPT², scFoundation³, Geneformer⁴), and no claim about that class is drawn from them. An
883 earlier draft asserted that they did stand in for it; that is inconsistent with the algebraic scope of the
884 Gate-1 result and with limitation 1, and it is withdrawn.

885 **How a predicted effect becomes a predicted population.** A distributional score ranks *populations*,
886 so a predict-then-rank baseline must turn a predicted effect into a population, and that synthesis step

887 silently determines what structure the population can possibly contain. We therefore state it explicitly.
 888 A predicted population is built by applying the predicted effect to the query context’s *real control cells*,
 889 one cell at a time. This is what the latent-arithmetic models in this class do (encode observed cells, shift
 890 each cell’s code, decode each cell), and it is the only construction under which a predicted population can
 891 carry the control population’s covariance and any multimodality it has. The alternative and superficially
 892 natural construction, drawing each predicted population as control mean + δ + iid $\mathcal{N}(0, \sigma)$, must not be
 893 used for structure diagnostics: that cloud is unimodal by construction (isotropy index 0.998), so a k -
 894 means $k=2$ between-total variance ratio computed on it returns ≈ 0.009 whatever the predictor learned.
 895 Any structure diagnostic evaluated on such a synthesizer measures the synthesizer, not the predictor.
 896 Both constructions are implemented behind an explicit flag so the difference can be reproduced.

897 **In-sample ceiling.** In the predict-then-rank experiment the predictors are fit on the full SciPlex3
 898 matrix and the query is composed of cells from two cell lines that remain in the training set, so the
 899 nearest-neighbour predictor can in principle transfer a signature it estimated from the very cells that
 900 compose the query. The reported predict-then-rank numbers are therefore an in-sample ceiling, not a
 901 transfer result. A leave-both-contexts-out option is provided, and we do not claim a leave-one-context-
 902 out guarantee for the reported run. The direction of this bias favours the mean incumbent, so it does not
 903 manufacture the negative result reported here; it bounds it.

904 **Gate 1 measured within a context, and the calibration check that caught us**

905 **The theorem is a within-context statement.** An additive predictor adds one delta vector to every cell
 906 of the query context. Its two subpopulations therefore receive identical deltas and $\cos(d_0, d_1)$ is exactly 1,
 907 but only if the two subpopulations are in that context. Our first Gate-1 experiment measured the cosine
 908 on an alpha-blended candidate population whose two subpopulations are two different cell lines, and
 909 an additive predictor emits a different delta per cell line, so what it measured was $\cos(\delta_{A549}, \delta_{K562})$. That
 910 is the similarity of two contexts’ deltas and is not a test of additivity.

911 **The calibration check is what exposed it, and it is asserted rather than inspected.** The experiment
 912 scores predictors that are additive *by construction*, whose value is fixed by the algebra at 1.000. They
 913 returned 0.267 and 0.353. A control that is required to hit a known value and misses it condemns the
 914 instrument, not the models, and the script refuses to emit numbers when it fails. We recommend the
 915 practice generally: an experiment whose theory pins a control to an exact value should assert that value,
 916 not merely report it.

917 **The corrected design** (analysis/predictors/gate1_within_context.py). For each context we take up
 918 to 400 control cells and split them into two subpopulations by their own cell state (30-component PCA,
 919 k -means with $k = 2$; subpopulation sizes and silhouettes are reported, because a low induced cosine
 920 would mean nothing if the within-context structure were not there). For each drug the predictor is run
 921 separately on each subpopulation’s control cells, so the partition is true by construction and no assignment
 922 step is involved:

$$d_k = \text{mean}(\text{predicted cells of subpop } k) - \text{mean}(\text{control cells of subpop } k), \quad \cos(d_0, d_1).$$

923 Population synthesis draws a permutation of the supplied cells, which preserves a mean exactly. The
 924 real treated cells of the same context, partitioned by the same control-derived subpopulations, give the

reference value. Gate 2's assignment error is no longer entangled with this measurement; it is measured separately (above).

Two baselines, because a generative model does not return $C + \delta$. A latent predictor returns $P(C) + W\delta_z$, where P is its own reconstruction of the input. The response of a subpopulation therefore depends on what it is referred to, and the two choices are not the same quantity:

$$\underbrace{d_k = \text{mean}(P(C_k)) - \text{mean}(C_k) + W\delta_z}_{\text{against the REAL control cells: what a scorer measures}}, \quad \underbrace{d_k = W\delta_z}_{\text{against the model's own decoded control}}.$$

The first carries $\text{mean}(P(C_k)) - \text{mean}(C_k)$, the reconstruction bias of subpopulation k , which is cell-state dependent and does not cancel. The second cancels it exactly. We report both. The additivity theorem is a statement about the second, and the calibration check is asserted on it: an additive predictor must return 1.000000, and a run in which it does not is discarded. A predictor that does not reconstruct its input (the gene-space average-effect predictor, and the optimal-transport map, which returns C plus a per-cell displacement) has no baseline of its own, and for those two the columns coincide by construction, which is the correct answer rather than a bug. Which predictors reconstruct and which do not is declared explicitly in the code rather than inferred.

Information-condition diagnostics and gate diagnosis

The information condition is tested as two gates. Gate 1 (differential response) uses the predict-then-rank experiments: candidate responses are generated by each predictor family, ranked by DART, and compared to mean retrieval, with structure diagnostics computed on generated versus real populations. Two quantities are reported. The *subpopulation-variance ratio* is the k -means $k=2$ between-total variance ratio of the candidate population. The *induced response cosine* is $\cos(d_{\text{maj}}, d_{\text{min}})$, where d_k is the mean of subpopulation k minus the mean of its own matched control population; it is the quantity a distributional score must exploit, and it is identically zero for any predictor that adds a single delta vector to every cell. The reference population for both is the alpha-blended candidate population that a scorer actually ranks, not a single-context treated population.

Gate 2 (recoverability) asks whether a query's subpopulation structure can be resolved at all, on a controlled mixture with known two-state composition (K562 at 10 μM , HDAC-class versus JAK-class cells; 50–400 cells per source, 10 seeds). The design point is that clustering failure on its own is uninformative: it is equally consistent with the information being absent from the representation and with its being present but unreachable without labels, and those have opposite implications for whether single-cell resolution can ever pay off. The analysis therefore reports a supervised ceiling alongside the unsupervised methods, and the decisive quantity is the *gap* between them, not either number against chance.

Common unit. Adjusted Rand index and classifier AUC are not comparable, so every method is converted to best-permutation accuracy against the true labels: each cluster is assigned its majority true class. For $k = 2$ this is the better of the two permutations; for $k > 2$ it is strictly generous, since an over-partitioning method is never penalised for splitting a true class. HDBSCAN noise is treated as its own cluster and likewise assigned its majority class, which is generous again. Clusterers thus receive their label matching for free.

Unsupervised methods: k -means and Gaussian mixtures, both handed the true $k = 2$; Leiden (graph-

based, resolution 1.0, which selects its own k); HDBSCAN (density-based, which selects its own k and may return none). The last two are the field’s actual defaults and were absent from our earlier analysis; adding them does not change the answer. All four see the same PCA representation (30 components) fit on the full mixture, which is the transductive setting an unsupervised pipeline would in fact have.

Supervised ceiling: logistic regression, random forest and k -nearest-neighbours, each given the ground-truth labels under 5-fold stratified cross-validation, with the PCA *re-fit inside each training fold* and test cells only projected through it, so no test-fold covariance informs the representation. The ceiling per run is the best of the three; including a nonlinear learner is what distinguishes an information limit from a too-weak-probe artifact.

The direction of the resulting bias, corrected. An earlier version of this Methods section stated that giving the clusterers the easier transductive representation while making the probe re-fit its PCA in-fold biased the comparison *against* the conclusion drawn. That is backwards. The conclusion drawn is that the *gap* is small; inflating the unsupervised arm and deflating the supervised arm makes the gap *smaller*, so the design flatters that conclusion rather than challenging it. We therefore also report the ceiling under the *matched* representation, the same transductive PCA the clusterers receive, which leaks no labels because PCA does not see them, and it is the matched gap that should be read (analysis/natural/gate2_drug_response.py).

Positive control. Source separation is scaled by a factor $s \in \{1, 1.5, 2, 3, 5, 8\}$ using a transform fit once on the full source populations, never on the sampled cells; $s = 1$ is the real data. Verifying that the ceiling pulls away from clustering as s rises is what licenses reading the absence of a gap at $s = 1$ as a measurement rather than as an insensitive probe.

Query identifiability on real drugs is the silhouette of a two-way k -means split computed per drug on the query cells alone.

The gate is additionally audited directly by re-running the partial-observed retrieval with an additively computed true response divergence column (leave-out-free, from the observed candidate populations), then correlating each gate axis with true divergence and stratifying retrieval gaps by divergence quartile with Benjamini–Hochberg control and power analysis. This audit is the basis for the findings that the structure-reliability axis is anti-correlated with true divergence and that the minority-coverage effect, while significant, is negligible.

Dataset positions on the divergence gate. For each dataset the response cosine between the two subpopulations of a query is computed from the observed populations with the same probe used inside the gate, giving 0.032 (cross-line SciPlex3), 0.709 (Frangieh) and 0.186 (CD34+).

HIR-Bench generator and synthetic welfare oracle

HIR-Bench generates heterogeneous retrieval tasks from a specified two-subpopulation latent welfare model with tunable conflict, yielding an analytical flip boundary $\alpha^* = B/(A + B)$. Each task is instantiated under the observed, predicted_mean, and predicted_structure information conditions. Reported numbers come from the full 13,440-*instance* grid (672 parameter configurations $\times$ 20 seeds; $n_+ = 8,640$, $n_- = 4,800$).

The predictability model. A logistic regression predicts retrieval flips from two feature sets, fitted and reported side by side because the contrast between them is the result. The *observable* set contains only quantities a retrieval method can compute at query time: the subpopulation-variance ratio, isotropy

index and response diversity of the query population and of the candidate populations. The *oracle-derived* set contains top- k disagreement and Kendall conflict scores, each of which is a function of the benchmark’s latent utility matrix, as is the label.

Folds are held out one *label-determining parameter cell* at a time, that is, one $(\alpha, \text{conflict})$ pair, because the label is a deterministic function of those two axes; there are 28 such cells and this is the effective sample size. Grouping by the instance identifier instead, the natural choice, is invalid here: that identifier also encodes axes that leave the utility matrix unchanged, so each held-out instance leaves roughly twelve near-identical copies of itself in the training set. We report the AUC under that grouping (0.640 for the oracle-derived features) only to quantify the inflation it produces. Because the label is a deterministic step function of the grouping variables, every held-out fold is single-class, no per-fold AUC is defined, and the pooled out-of-fold AUC is the only available statistic.

Partial-observed retrieval protocol and real-data projection

The partial-observed protocol scores candidates by an energy-based welfare regret against the latent objective and records both the objective-aligned regret (Class A) and task-proximal metrics (Class B: MoA-recovery nDCG, minority-state coverage) per query.

Held-out unit, and what it does not license. The held-out unit in the gate audit is a cell line (A549, K562, MCF7), and all three come from SciPlex3, so they share batch, protocol, highly-variable-gene selection and normalization. This is leave-one-cell-line-out within a single experiment, not leave-one-dataset-out, and we make no cross-dataset generalization claim on its basis.

Undefined metric values are excluded, not encoded. MoA recovery is *undefined* for the leave-MoA-out and partial-library splits, where the entire mechanism class is removed from the candidate library, so no correct answer is present to recover. Those 165 of 765 queries (21.6%) are dropped, giving $n = 600$. They must not be carried as a sentinel value: differencing a sentinel against itself yields a structural zero that enters every stratum as a spurious no-effect observation and pins the median gap at exactly 0.000. Excluding them does not rescue the metric; it makes the negative result sharper.

Real-data projection. The projection maps the four datasets onto the HIR-Bench regime plane and records, per task, the minority-state coverage of the DART pick minus that of the mean pick. The full configuration builds 239 tasks, and the composition is given here because the three-factor description in an earlier draft accounted for only 180 of them: 90 constructed cross-line tasks (three cross-line pairs $\times$ ten drugs $\times$ three seeds), 90 within-line (three cell lines $\times$ ten drugs $\times$ three seeds), 23 Frangieh, 12 CD34+ and 24 under predicted candidates. We report the distribution of that difference.

Why the dominance count is withheld. A task is conventionally called distributionally dominant when the difference exceeds $+0.01$. We compute the count and do not report it as a finding, for three reasons. (i) The threshold is not calibrated to the data: it sits at 0.77 standard deviations of the nonzero-difference distribution (s.d. = 0.013), and it is one-sided, so on the full run 11 tasks clear it for DART while 2 clear it for the mean and only the first number would conventionally be quoted. (ii) The seed is not a replicate. It re-draws which drugs are held out and it redefines the minority subpopulation by re-clustering, so a task recorded as dominant at one seed carries no information about robustness. (iii) When the drug is held fixed and the seed is resampled ten times, only 2 of the 10 threshold-crossing cross-line tasks remain above threshold in at least 8 of 10 seeds, while control drugs that cross it at no seed in the recorded run cross it in 1 to 4 of 10 resamples. A control here is a drug that is dominant at no

recorded seed; selecting controls by task identifier alone would let a drug that crosses at one seed and misses at another fall into both groups. The count is a count of noise excursions. The distribution is not, and it is what we report.

Two caveats bound the projection. First, the regime boundary it transfers is fitted on HIR-Bench’s method-performance layer, which is available on a 144-instance grid because scoring nine retrieval methods against up to 100 candidates per instance does not scale to the full grid. The boundary is therefore fitted on 72 grid cells, and we claim no more precision for it than that supports. Second, the projection’s own acceptance test fails. Its label derives from the same threshold we decline to trust, and it is heavily imbalanced (228 to 11), so a constant predictor scores 95.4% while the regime projection scores 61.1%, which is 34 points *worse* than saying nothing. The HIR-Bench boundary does not transfer to real data, and that holds whatever one makes of the threshold.

Evaluation metric taxonomy and circularity audit

We classify every evaluation metric into three classes. Class A: objective-aligned proxy metrics measure the same mathematical object the method optimizes or scores (energy-distance regret for energy-based retrieval, connectivity score for signature retrieval, reconstruction error for expression predictors, graph proximity for graph inverse design); they test objective fidelity, not independent utility. Class B: task-proximal metrics are biologically closer but still derived from the same upstream data (MoA recovery, target-family recovery, DEG overlap, pathway enrichment, minority-state coverage). Class C: oracle-independent utility metrics have external functional grounding (dose-response viability, longitudinal survival, post-treatment clone expansion, validated resistance emergence, functional readouts, combination efficacy) and are required for therapeutic-utility claims.

A field-level audit across six published method families, spanning signature retrieval^{8–10}, expression predictors and foundation models^{2–6,13}, and graph/target inverse design^{7,14,15}, finds that every family’s *headline* metric falls in Class A and that none reports a Class-C readout as its primary evidence; population-level distributional metrics are themselves standard in recent perturbation-prediction benchmarks¹². The audit, its inclusion criteria, and the rule by which alignment risk is judged are given in full as Supplementary Table 2, so the claim can be checked rather than taken on assertion. It is an audit of *evaluation practice*, not of correctness, and the family is the unit: we do not score individual papers. We report DART under all three classes. No Class-C readout exists within the single-cell datasets themselves; the Class-C analysis below therefore imports one, and we label it semi-real for that reason.

Class-C analysis: an independent functional oracle and its scalar controls

Choosing the oracle. A retrieval score is asked which candidate’s response distribution most resembles the query’s. It is not asked which candidate is most cytotoxic. Grading it on *absolute potency* therefore poses a question the method was never set, and it does so in a way that is not neutral: potency is largely driven by response magnitude, so a scalar that sorts candidates by magnitude is near-guaranteed to win, and a similarity retriever handed a weak query will correctly return other weak drugs and be scored as if it had failed. We therefore take the *semantically matched* oracle to be drug–drug functional similarity and report the absolute-potency comparison only as a confounder audit.

Construction. SciPlex3 drugs were matched to GDSC records by normalized drug name (exact match, substring matching as a fallback), giving 35 matched compounds after removing one duplicate

mapping (two SciPlex3 entries resolving to the same GDSC compound) and restricting to a *single* screening platform, GDSC2. That platform restriction matters: GDSC1 and GDSC2 AUCs come from different assays and dose ranges and are not on a common scale, so pooling them would inject an assay-of-origin step into the oracle itself. For each matched compound we take its dose-response AUC profile across GDSC2 cell lines and define

$$\text{oracle}(q, c) = \text{Spearman}(\text{AUC}_\bullet(q), \text{AUC}_\bullet(c)),$$

the rank correlation of the two compounds’ cross-cell-line sensitivity patterns. Three properties make this a usable Class-C oracle. It is *query-dependent*, so a query-independent scalar cannot game it. It is *out of context*: the three SciPlex3 lines (A549, K-562, MCF7) are *excluded* from the profile, leaving 966 cell lines, so the oracle is never evaluated in the biological context the transcriptional data came from. And because a correlation is centered per compound, a drug’s overall potency *level* cannot drive it; only its selectivity *pattern* can, which is precisely the channel the absolute-potency comparison confounds.

Cell lines differ greatly in general drug sensitivity (mean AUC across all GDSC2 compounds ranges from 0.720 to 0.966), and that shared main effect inflates every drug–drug correlation: left uncorrected, 85% of drug pairs appear positively similar (median +0.178). We therefore subtract each cell line’s mean AUC, estimated across all 286 GDSC2 compounds rather than only the 35 matched ones, before correlating; the corrected oracle is symmetric about zero (52% positive, median +0.015). The correction was specified on this reasoning before the retrieval results were inspected. Both oracles are reported: they change every scorer’s absolute value but not the ordering that carries the conclusion. Pairs sharing fewer than 100 cell lines are dropped, leaving 591 usable drug pairs at a median of 909 shared lines.

Task. Per cell line, leave-one-drug-out: the held-out drug’s 10 μM treated population is the query and the remaining matched drugs are the candidates (103 queries; 34–35 candidates per line). Each rule ranks the candidates and the ranking is compared with the oracle by Spearman correlation. The oracle is visible to no scorer.

The rankings and their controls. (i) *energy*, the distributional score. (ii) *mean cosine with control subtraction*, the correctly specified incumbent (treated mean minus matched control mean). (iii) *mean cosine without control subtraction*, how a mean baseline is often run in practice, which measures baseline cell identity rather than drug response. (iv) *magnitude matching*: rank by $-|\|d_c\| - \|d_q\||$, preferring candidates whose response is about as large as the query’s. This is query-dependent yet compares no distributions whatsoever, and it is the null a distributional score must beat. (v) *magnitude only*: rank by $\|d_c\|$, which is query-independent and produces one ordering for every query; on a query-dependent oracle it is a falsifiable sanity check rather than a competitor. (vi) *potency matching*: rank by $-|\text{AUC}_c - \text{AUC}_q|$ in the query’s own line. This is a *diagnostic*, not a deployable recommender, since it requires the query drug’s own measured potency; it tests how far the functional oracle is re-encoding potency level, and we report that it does so substantially (median $\rho = +0.399$).

Reported statistics. Per-query Spearman correlations summarized by their median. Energy is additionally partialled on the magnitude-matching channel by rank-residualizing both the score and the oracle on it, which isolates the contribution attributable to the distribution rather than to magnitude. The 103 queries within a cell line share a candidate pool, so paired tests across them are anticonservative; we report such a p -value once, explicitly flagged, and treat the three cell lines as the honest unit, giving per-

line medians throughout. We do not convert any rank correlation into a fraction of variance explained; a Spearman ρ is not one.

Confounder audit (absolute potency). The same rankings scored against absolute GDSC2 AUC are reported in Extended Data. There, energy is anti-correlated with potency ($\rho = -0.520$) and the query-independent magnitude scalar reaches $+0.692$. This follows arithmetically: response magnitude predicts potency ($\rho = -0.58$ to -0.71 by line), and the energy *distance* between query and candidate largely tracks the candidate’s own magnitude ($\rho = +0.791$), so ranking nearest-first returns the smallest-response candidates. We report this to expose the magnitude channel, and we draw no conclusion about therapeutic utility from it.

The oracle-shape test: holding the data fixed and changing only the judge

Data. Frangieh Perturb-CITE-seq measures RNA and 20 surface proteins (24 antibodies, 4 isotype controls removed) in the same cells. The two modalities are joined by cell barcode, not by row position: the RNA and protein matrices as distributed differ in row count, so a positional concatenation would pair each cell’s protein response with a different cell’s RNA, which produces a result that looks unremarkable and is pure noise. The join is asserted, giving 218,331 cells. Protein counts are CLR-normalized across antibodies per cell. Analyses are run *per immune condition* (Control, IFN γ , Co-culture; these live in `perturbation_2`, not `condition`), because pooling them confounds condition with perturbation. Knockouts with at least 60 cells are used, subsampled to 250, seed fixed.

The two oracles. Both are built from the *same* protein measurements of the *same* cells, and differ only in statistical form:

$$\text{oracle}_{\text{MEAN}}(q, c) = \cos(\bar{d}_q^{\text{prot}}, \bar{d}_c^{\text{prot}}), \quad \text{oracle}_{\text{DIST}}(q, c) = -\mathcal{E}(D_q^{\text{prot}}, D_c^{\text{prot}}),$$

where $\bar{d}_k^{\text{prot}}$ is the mean protein response of knockout k against that condition’s control mean, D_k^{prot} is the set of *per-cell* protein responses of knockout k against the same control mean, and $\mathcal{E}$ is the energy distance. Centering both on the control is what makes the distributional oracle a *response* distance rather than a baseline-identity distance.

What is held fixed. The two RNA rankings under test, energy and control-subtracted mean cosine, are computed once and scored against both oracles, so they are identical across the two columns by construction. No scorer sees protein data of any kind. Nothing but the oracle’s form changes.

The magnitude control, and why it is mandatory here. Both distributional objects in this design are energy distances, and an energy distance largely tracks a candidate’s own response magnitude ($\rho = +0.791$; Extended Data Fig. 5). A magnitude-to-magnitude channel could therefore reproduce the reversal without any distribution being compared, and a paper that requires this control of others cannot omit it from its own sharpest claim. We therefore add a third RNA ranking, magnitude matching ($-|\|d_c\| - \|d_q\||$), which is query-dependent and compares no distributions, and score it against both oracles. Reported statistics are per-query Spearman correlations summarized by their median, with the three immune conditions given separately as the honest unit.

1158 Resistance-associated exploratory analysis

1159 Response divergence per cell is correlated with curated resistance-associated program scores
1160 (AXL/mesenchymal, interferon-response, antigen-presentation) in the Frangieh data. This analysis is
1161 exploratory; no survival, timecourse, or functional readout is available, and no resistance-prediction
1162 claim is made.

1163 Natural heterogeneity: patient-derived glioblastoma

1164 **Data.** ZhaoSims2021: acute-slice culture and surgical biopsy from ten glioblastoma patients, treated
1165 with six compounds (Ana-12, Ipsenib, RO4929097, Tazemetostat, etoposide, panobinostat). Counts
1166 are normalized to 10^4 , log1p-transformed, and reduced to 2,000 highly variable genes. This is not a
1167 retrieval benchmark and is not used as one. Only one patient received more than two compounds, so a
1168 leave-one-drug-out retrieval would run on a single patient against five candidates; reporting Hit@1 from
1169 that would be an anecdote presented as a benchmark. It is used to test the two gates, and the paper’s
1170 motivating premise, on heterogeneity that nobody constructed.

1171 **Compartment assignment, and why it is not an argmax.** The published annotation carries celltype
1172 = unknown for every cell, so compartments are assigned by canonical marker score: each cell receives
1173 a z-scored mean expression for each marker set, computed on the full log-normalized matrix before
1174 highly-variable-gene selection so the labelling never depends on a gene set chosen for downstream vari-
1175 ance. A first version of this procedure took the bare argmax over five marker sets, which *forces* every cell
1176 into some compartment including cells that express no markers at all. It produced two compartments
1177 that fail their own validation: cells labelled T cells (22% of the dataset) had a mean CD3D/CD3E/CD2
1178 expression of 0.080, *lower* than their malignant-marker score of 0.141, and cells labelled endothelial ex-
1179 pressed myeloid markers more strongly than endothelial ones. They were neither; they were unidenti-
1180 fiable cells swept into whichever class the argmax had left over. Building a natural-heterogeneity result
1181 on a fabricated T-cell compartment would be worse than having no natural dataset.

1182 A cell is therefore assigned to a compartment only if it scores highest for that compartment, clears
1183 an absolute expression floor on that compartment’s key markers (≥ 0.25 mean log-normalized expres-
1184 sion), and beats the runner-up by a z-score margin (≥ 0.25). Everything else is labelled unassigned and
1185 *excluded*, not reassigned; 39.9% of cells are dropped this way. The natural-tissue result is insensitive to
1186 these two constants. Over floor = margin $\in \{0.10, 0.25, 0.40, 0.50\}$, which drops between 29% and 58%
1187 of cells and passes compartment validation at every setting, the Gate-2 ceiling stays 0.915–0.923, the
1188 best unsupervised method 0.77–0.80, the gap 0.11–0.13, and the Gate-1 induced cosine 0.54–0.59 (analy-
1189 sis/natural/zhao_threshold_sensitivity.py). The 0.25 setting is neither the most nor the least favourable.
1190 Only the three compartments that pass validation are used: malignant glioma cells (own-marker expres-
1191 sion 1.02 versus 0.07 and 0.12 for the others), tumour-associated myeloid cells (1.53 versus 0.12, 0.17)
1192 and oligodendrocytes (1.78 versus 0.41, 0.11). The loader raises rather than writing labels if any called
1193 compartment fails to express its own key markers most strongly, and the validation table ships with the
1194 results. Labels are never shown to any clustering method or to any retrieval score; they exist only to
1195 evaluate whether those methods recover them.

1196 **Gate 1 (natural).** For each (patient, drug) with at least 50 cells in both the malignant and the myeloid
1197 compartment, $d_k = \text{mean}(\text{treated cells of compartment } k) - \text{mean}(\text{that patient’s control cells of compartment } k)$,

and the reported quantity is $\cos(d_{\text{malignant}}, d_{\text{myeloid}})$. Controls are matched *within patient and within compartment*, so the cosine cannot be inflated by a shared patient or compartment effect.

Gate 2 (natural), and the comparison our first attempt got wrong. Gate 2 asks whether the subpopulations *inside a candidate population* can be told apart, which is a *drug-response* partition. Our first natural test instead separated malignant from myeloid *control* cells, a *cell-type* partition between two lineages that differ in thousands of genes. It is a far easier problem, it returned a ceiling of 0.964, and comparing that with the constructed mixture's 0.692 compared two different questions while holding the protocol fixed. That number is withdrawn.

The reported analysis (analysis/natural/gate2_drug_response.py) poses the same question in both settings. A split is a pair of drugs: cells treated with drug A against cells treated with drug B, *within one compartment and within one patient*, at least 50 cells per drug, capped at 150 per source, 5 seeds. This gives 36 drug-pair splits over 4 patients and 2 compartments. The constructed comparator is run at the same granularity, one drug against one drug in K562 at 10 μM (24 HDAC-versus-JAK drug pairs, subsampled with a fixed seed from the 513 available and reported as subsampled). The class-pooled split the paper previously used is retained as a third arm, because a pooled class carries the extra within-class variance of several distinct compounds and is therefore harder for a reason that has nothing to do with the mixture being constructed. Quoting it against the natural arm would inflate the contrast.

The representation, and a bias we had stated backwards. Everything else follows the constructed-mixture protocol: a common 30-component PCA for the unsupervised methods, best-permutation accuracy as the shared unit with label matching given to each clusterer for free, and a supervised ceiling taken as the best of logistic regression, random forest and k -nearest-neighbours under 5-fold cross-validation. We report the ceiling under two representations. The *in-fold* ceiling re-fits PCA inside each training fold. This is the leak-free choice, but it is not the representation the clusterers were given; because it lowers the ceiling it *shrinks* the gap, which is the direction that supports the conclusion the gap is used to draw. The *matched* ceiling uses the same transductive PCA the clusterers receive, which leaks no label information because PCA never sees labels, and it is the apples-to-apples quantity; it is what we report. *The reported gap is the median of the paired per-split differences*, not the difference of the two marginal medians; a median is not linear and the two disagree (for the natural arm, +0.117 against +0.147), and the paired statistic is the correct and the smaller one.

Uncertainty on the gap (analysis/natural/gate2_uncertainty.py). Three things make the raw number fragile and each is addressed. (i) The gap is a difference of two *maxima*, best of three probes minus best of four clusterers, and a maximum over methods is upward-biased. We therefore also compute every fixed (probe, clusterer) pair, in which no selection occurs; HDBSCAN is excluded from the summary because it assigns every cell to noise and scores exactly 0.500, so a gap measured against it reflects a clusterer that did nothing rather than information that is unreachable. (ii) The confidence interval is a cluster bootstrap over patients, not over splits. Thirty of the 36 natural splits come from one patient and reuse the same six drugs, so a split-level bootstrap would treat dependent observations as independent and return an interval several times too narrow, which is precisely the pseudo-replication this paper criticises elsewhere. With only four clusters the bootstrap is coarse and is known to under-cover, so the interval is reported as a floor on the uncertainty rather than a faithful estimate of it. (iii) The dominant patient is dropped entirely and the analysis repeated on the remaining three.

1239 **The premise.** For every pair of drugs within a patient, we compute the cosine between their pop-
1240 ulation *mean* signatures and the cosine between their *malignant-compartment* responses, and report the
1241 Spearman correlation between the two across drug pairs. If the mean were a sufficient statistic these
1242 would be the same number. *No pass/fail threshold is applied.* An earlier version of the analysis script de-
1243 clared the premise supported if the correlation fell below 0.9, and the observed value was 0.878. That
1244 threshold was ours, it was applied after seeing the number, and clearing it by 0.02 licenses nothing; the
1245 verdict has been removed from the code and the correlation is reported and interpreted in the text.

1246 **Identifiability phase diagram**

1247 The phase diagram sweeps a separation scale and a cell budget on the K562 HDAC-versus-JAK mixture,
1248 whose labels are known by construction. Separation is applied by pushing each source along the vector
1249 from its centroid to the global centroid. Those centroids are computed once, on the *full* source popula-
1250 tions, so the manipulation is independent of the sample being evaluated. This is not a detail: computing
1251 the centroids on the sampled cells and then shifting those same cells fits the manipulation on the data it
1252 is measured on, amplifies whatever separation each small draw happens to contain, and produces the
1253 indefensible result that the adjusted Rand index *falls* as the cell budget rises. Clustering does not become
1254 harder with more cells.

1255 **Statistical analysis**

1256 Paired comparisons use the Wilcoxon signed-rank test; group comparisons use Mann–Whitney U;
1257 monotone associations use Spearman correlation. Multiple comparisons across strata use Benjamini–
1258 Hochberg false-discovery-rate control, and we report achieved statistical power and the sample size
1259 required for 80% power where effects are small. Effect sizes are reported alongside *p*-values throughout,
1260 and statistically significant but negligible effects are labeled as such. Where an effect is reported as
1261 negligible, the operating range of the metric is given so that the effect size can be read against it.

1262 **Data availability**

1263 All datasets analysed in this study are publicly available.

1264 Three of the four single-cell datasets were obtained in harmonized form from scPerturb¹⁰, Zenodo
1265 record 13350497 (DOI [10.5281/zenodo.13350497](https://doi.org/10.5281/zenodo.13350497), release 1.4), and we name the exact file for each so the
1266 analysis can be reproduced from the same bytes rather than from a re-derivation of the primary data.
1267 SciPlex3: file SrivatsanTrapnell2020_sciplex3.h5ad; primary data in Gene Expression Omnibus un-
1268 der accession GSE139944²⁴. Frangieh melanoma Perturb-CITE-seq: files FrangiehIzar2021_RNA.h5ad
1269 and FrangiehIzar2021_protein.h5ad (the protein modality is used by the surface-protein oracle); orig-
1270 inal data²⁵. ZhaoSims2021 patient-derived glioblastoma (the natural-heterogeneity benchmark): file
1271 ZhaoSims2021.h5ad; original data²⁶. CD34+ HSPC: Gene Expression Omnibus, accession GSE306429.

1272 **GDSC dose-response** (the Class-C functional oracle): Genomics of Drug Sensitivity in Cancer,
1273 GDSC2 fitted dose-response, release 8.5 (file GDSC2_fitted_dose_response_270ct23.xlsx), available
1274 from <https://www.cancerrxgene.org> and the Sanger bulk-download mirror. The specific release was
1275 recovered and verified: the retained join table (drug_match_table.csv) reproduces the AUC and ln IC₅₀
1276 of this release exactly across all checked (compound, cell line) pairs, and the Class-C statistics were re-
1277 derived from it to confirm they are unchanged. The join table is released with the code so the drug

1278 matching can be audited independently.

1279 **LINCS L1000 / PDGrapher benchmark:** a precomputed benchmark table ships with the code
1280 (`pdgrapher_closed_loop_benchmark.parquet`). No reported result in this paper derives from it, so it
1281 is not data underlying the findings; we name it only because it is present in the release, and we do not
1282 have a retained accession for it.

1283 The processed tensors used by the experiments, together with the scripts that build them from the
1284 primary data, are described in the repository. Source data for every figure panel are provided as the
1285 experiment output tables in the code repository.

1286 Code availability

1287 All code, experiment scripts, and the evaluation-metric taxonomy are available at [https://github.com/](https://github.com/Boom5426/DART)
1288 [Boom5426/DART](https://github.com/Boom5426/DART) (**[[archived DOI on Zenodo at acceptance]]**). The release includes a test suite and a nu-
1289 merical consistency check that re-derives every reported constant from the cached experiment outputs;
1290 the latter re-reads the cached experiment tables and diffs them against the recorded constants, and is a
1291 consistency check rather than an end-to-end re-execution. A single driver script re-executes the pipeline
1292 from the processed tensors.

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1295 Author contributions

1296 **[[Author contributions, CRediT taxonomy: conceptualization, methodology, software, formal analy-
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1298 supervision, funding acquisition.]]**

1299 Competing interests

1300 **[[Competing interests statement. If none: “The authors declare no competing interests.”]]**

1301 Additional information

1302 Supplementary Information is available for this paper: Extended Data Figs. 1–6, Supplementary Notes
1303 1–3 and Supplementary Tables 1–4.

1304 Correspondence and requests for materials should be addressed to **[[Name]]**.

1305 References

- 1306 [1] C Bunne, Y Roohani, Y Rosen, et al. How to build the virtual cell with artificial intelligence: priori-
1307 ties and opportunities. *Cell*, 187:7045–7063, 2024. doi: 10.1016/j.cell.2024.11.015.
- 1308 [2] H Cui et al. scgpt: toward building a foundation model for single-cell multi-omics using generative
1309 ai. *Nature Methods*, 21:1470–1480, 2024.
- 1310 [3] M Hao et al. Large-scale foundation model on single-cell transcriptomics. *Nature Methods*, 21:1481–
1311 1491, 2024.

- 1312 [4] C V Theodoris et al. Transfer learning enables predictions in network biology. *Nature*, 618:616–624,
1313 2023.
- 1314 [5] M Lotfollahi, F A Wolf, and F J Theis. scgen predicts single-cell perturbation responses. *Nature*
1315 *Methods*, 16:715–721, 2019.
- 1316 [6] M Lotfollahi et al. Predicting cellular responses to complex perturbations in high-throughput
1317 screens. *Molecular Systems Biology*, 19:e11517, 2023.
- 1318 [7] Y Roohani, K Huang, and J Leskovec. Predicting transcriptional outcomes of novel multigene per-
1319 turbations with gears. *Nature Biotechnology*, 42:927–935, 2024.
- 1320 [8] J Lamb et al. The connectivity map: using gene-expression signatures to connect small molecules,
1321 genes, and disease. *Science*, 313:1929–1935, 2006.
- 1322 [9] A Subramanian et al. A next generation connectivity map: L1000 platform and the first 1,000,000
1323 profiles. *Cell*, 171:1437–1452, 2017.
- 1324 [10] S Peidli et al. scperturb: harmonized single-cell perturbation data. *Nature Methods*, 21:531–540,
1325 2024.
- 1326 [11] C Ahlmann-Eltze, W Huber, and S Anders. Deep-learning-based gene perturbation effect prediction
1327 does not yet outperform simple linear baselines. *Nature Methods*, 2025.
- 1328 [12] X Wei et al. Benchmarking algorithms for generalizable single-cell perturbation response prediction.
1329 *Nature Methods*, 2026.
- 1330 [13] L Hetzel et al. Predicting cellular responses to novel drug perturbations at a single-cell resolution.
1331 In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- 1332 [14] E Guney et al. Network-based in silico drug efficacy screening. *Nature Communications*, 7:10331,
1333 2016.
- 1334 [15] G Gonzalez et al. Combinatorial prediction of therapeutic perturbations using causally inspired
1335 neural networks. *Nature Machine Intelligence*, 2025.
- 1336 [16] C Bunne et al. Learning single-cell perturbation responses using neural optimal transport. *Nature*
1337 *Methods*, 20:1759–1768, 2023.
- 1338 [17] C Chi et al. Departures: distributional transport for single-cell perturbation prediction with neural
1339 Schrödinger bridges. *arXiv*, 2025. preprint arXiv:2511.13124.
- 1340 [18] Y Ji, T D Green, S Peidli, et al. Optimal distance metrics for single-cell RNA-seq populations. *bioRxiv*,
1341 2023. doi: 10.1101/2023.12.26.572833. preprint.
- 1342 [19] A Rubbi, A G Di Francesco, M Lotfollahi, and P Liò. A standardized framework for evaluating gene
1343 expression generative models. *arXiv*, 2026. preprint arXiv:2603.11244.

- 1344 [20] M Heidari, M Karimpour, S Srivatsa, and H Montazeri. Evaluating single-cell perturbation response
1345 models is far from straightforward. *bioRxiv*, 2026. doi: 10.64898/2026.02.14.705879. preprint.
- 1346 [21] D Agarwal and R Bisht. The metric picks the winner: evaluation choice flips model rankings for
1347 drug-response prediction in unseen chemistry. *arXiv*, 2026. preprint arXiv:2606.12639.
- 1348 [22] D Klein et al. CellFlow enables generative single-cell phenotype modeling with flow matching.
1349 *bioRxiv*, 2025. preprint 2025.04.11.648220.
- 1350 [23] K Z Kedzierska et al. Zero-shot evaluation reveals limitations of single-cell foundation models.
1351 *Genome Biology*, 26, 2025.
- 1352 [24] S R Srivatsan et al. Massively multiplex chemical transcriptomics at single-cell resolution. *Science*,
1353 367:45–51, 2020.
- 1354 [25] C J Frangieh et al. Multimodal pooled perturb-cite-seq screens in patient models define mechanisms
1355 of cancer immune evasion. *Nature Genetics*, 53:332–341, 2021.
- 1356 [26] Wenting Zhao, Athanassios Dovas, Eleonora F. Spinazzi, Hanna Mendes Levitin, Matei A. Bush,
1357 Peter Zhao, Steven S. Rosenfeld, Jeffrey N. Bruce, Peter Canoll, and Peter A. Sims. Single-cell analy-
1358 sis of glioblastoma tissue slices reveals drug-response heterogeneity across malignant and immune
1359 compartments. *Genome Medicine*, 13:82, 2021. doi: 10.1186/s13073-021-00894-y. **[[verify title and
1360 author list against the source before submission]].**